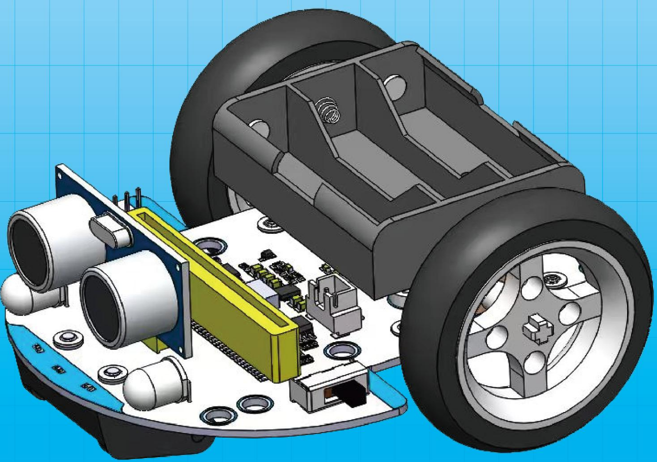
SIYEENOVE

SMART ROBOT CAR

User Guide

2025.9.14



POWERED BY MICRO:BIT

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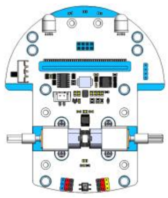
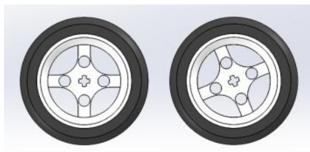
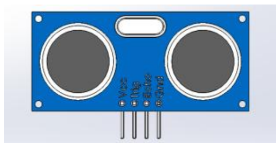
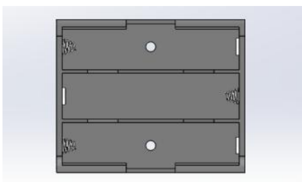
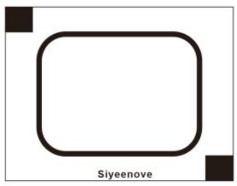
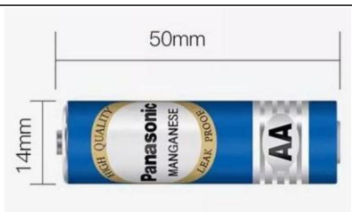
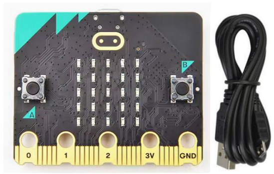
1. Introduction to Robot Car

This Micro:bit-based robotics kit integrates triple infrared line tracking, ultrasonic obstacle avoidance (P12/P13), and IR reception (P9). Features include RGB headlights and programmable RGB strips (P8) with triple servo drive ports. Expandable via I2C (P19/P20) and GPIO interfaces. Operates on 3-5.5V power with voltage indication. An ideal platform for learning robotic programming.

1.1 Product List

PCB base	Wheels	Ultrasonic module
		
AA Battery Case	Infrared remote control	Map
		
3M adhesive sticker		
		

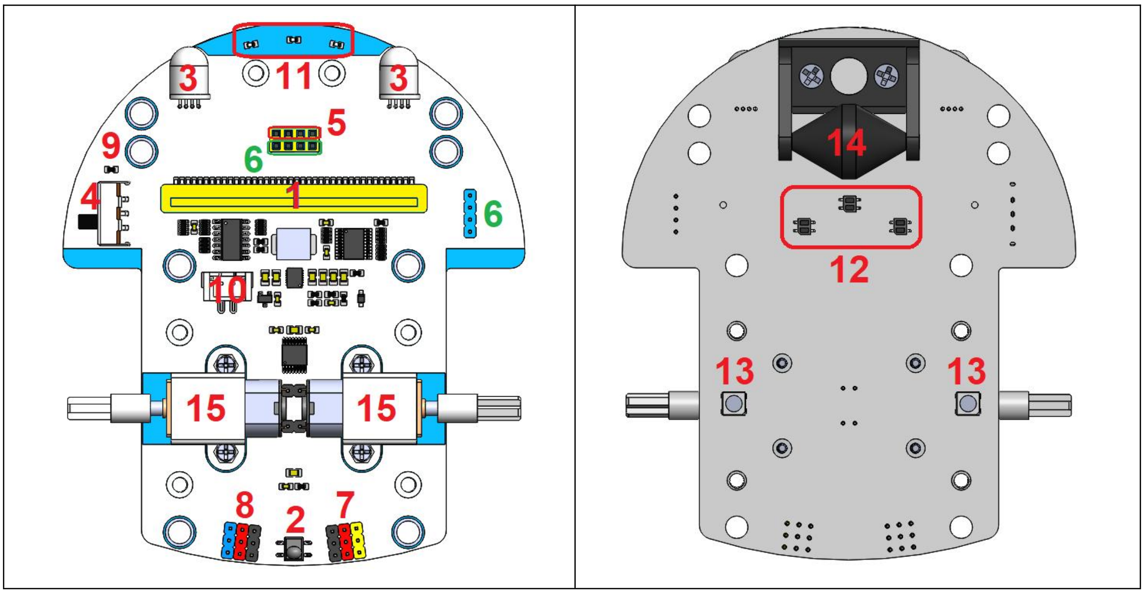
Note: You need to purchase three AA batteries and a Micro:bit v2 board with USB cable by yourself!

AA Battery	Micro:bit-v2 with USB cable
	

1.2 Specifications

Control board:	Micro:bit - v2 and v1 (not included)
Programming language:	MakeCode
Battery interface type:	XH2.54 2P, input voltage 3--5.5V, 3 AA batteries are required
Infrared remote control:	NEC, remote control distance ≧5 meters
Motor:	GA12-N20 micro DC gear reduction motor (145 rpm)
Ultrasonic sensor:	HC-SR04 (distance measurement: 2cm-400cm, accuracy: 3mm)
Extension interface type:	2.54mm pitch pin header
Wheel diameter:	53mm
Length and width:	117.5*97*70mm

1.3 PCB base introduction



1: Micro:bit slot

Can be plugged into a Micro:bit v2 board.

2: Infrared receiving sensor

Using NEC infrared communication protocol, its signal pin is connected to the P9 pin of Micro:bit.

3: Head light

5mm RGB LEDs (left/right) controlled by STC8H1K08 via Micro:bit's I2C commands.

4: Power switch

Controls the power on and off of the battery box.

5: Ultrasonic module interface

V=5V, G=GND, used to connect the HC-SR04 sonar module, the signal pins are connected to the P12 (Echo) and P13 (Trig) pins of Micro:bit respectively.

6: I2C port

It is the I2C communication interface of Micro:bit, SDA=P20, SCL=P19, which can communicate with other I2C devices.

7: Servo interface

Can be connected to 3 servos, which can be used to drive 90, 180, 270 and 360 degree servos.

8: IO expansion port

Can be connected to other sensor modules.

9: Power indicator light

After setting the battery model, this light will be on when the robot is working, and it will flash when the voltage is less than or equal to 20% of the voltage. (This only works if the battery type is set in the program)

10: Battery interface

XH2.54 2P, input voltage 3--5.5V.

11: Infrared line tracking indicator

The light turns on when the sensor detects a black line, and turns off when it detects a white area.

12: Infrared line tracking sensor

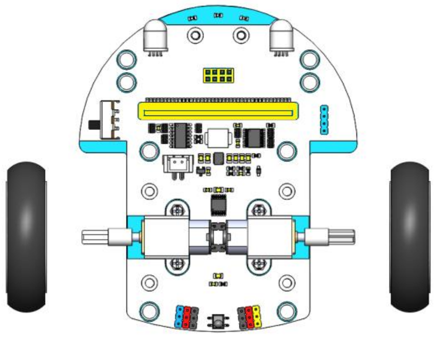
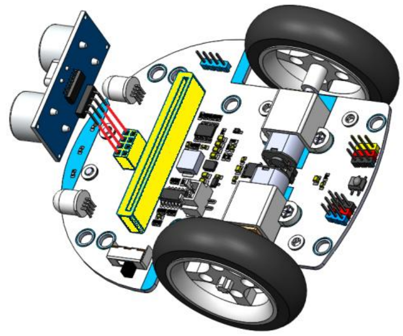
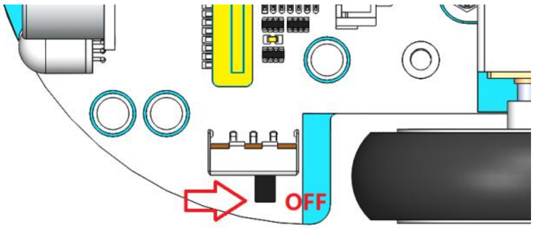
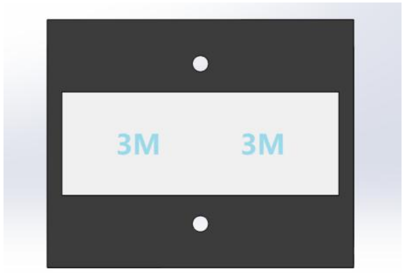
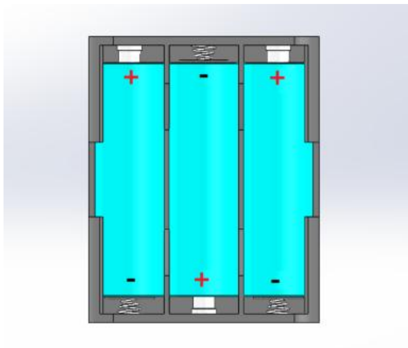
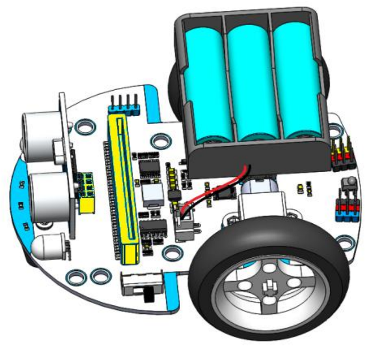
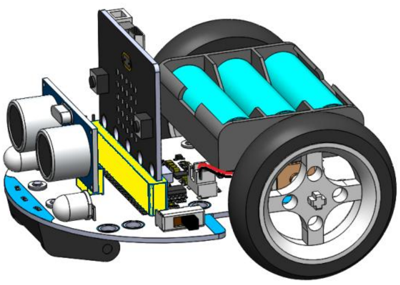
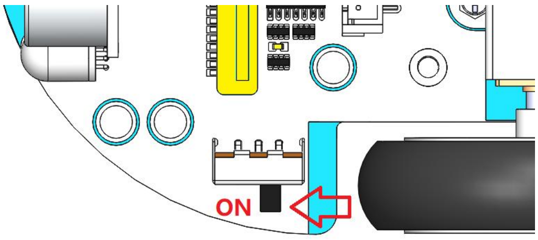
It is an infrared sensor that senses black and white areas through the principle of infrared reflection. There are three infrared line sensors on the left, middle and right bottom of the robot. The signal pins are connected to the P14, P15 and P16 pins of Micro:bit respectively.

13: RGB lights

Two WS2812 RGB lights at the robot's bottom produce mixed colors via individual R/G/B control, wired to Micro:bit P8.

14: steerable wheel **15: motor**

2. Assemble the Pybit

Step 1: Install the wheels	Step 2: Install the Ultrasonic Module
	
Step 3: Turn off the power(Note: Turn off the power when plugging or unplugging the Micro:bit V2 to avoid damaging it!)	Step 4: Stick the 3M double-sided tape on the center of the bottom of the battery case
	
Step 5: Install 3 AA Batteries	Step 6: Glue the Battery Box to the Robot
	
Step 7: Insert the Micro:bit V2 board (note that the dot matrix faces forward)	Step 8: Turn on the power switch
	

3. Getting Started with MakeCode

Microsoft's MakeCode editor is the perfect way to start coding with the BBC micro:bit. MakeCode is free and works across all platforms and browsers.

We recommend using Chrome or Edge browsers. WebUSB is a recent and developing web feature that allows you to access a micro:bit directly from a web page. It also lets you directly receive data into the MakeCode editor from the micro:bit. It works in Google Chrome and Microsoft Edge browsers.

WebUSB support for your micro:bit

If you're not using a current version of the Chrome or Microsoft Edge browsers, make sure they are this version or newer:

Chrome (version 79 and newer) browser for Android, Chrome OS, Linux, macOS and Windows 10.

Microsoft Edge (version 79 and newer) browser for Android, Chrome OS, Linux, macOS and Windows 10.

Link to download the latest Google Chrome:

https://www.google.com/chrome/

Link to download the latest Microsoft Edge:

https://www.microsoft.com/en-us/edge/download

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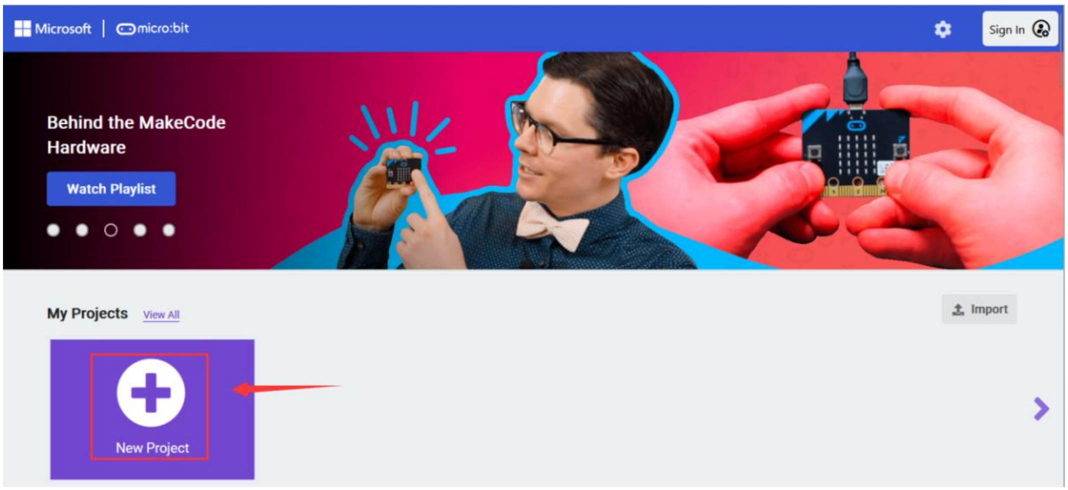
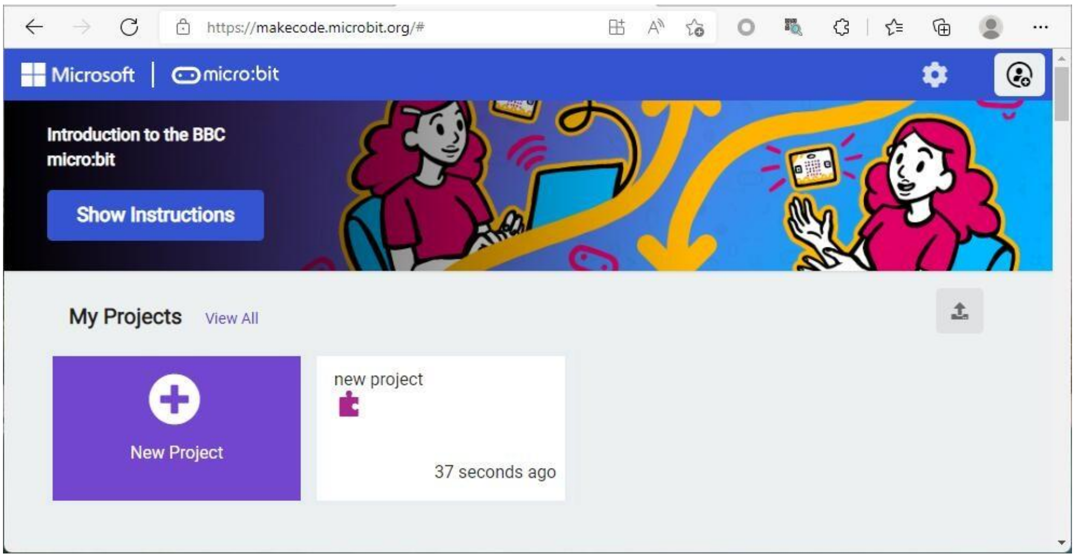
3.1 Create a new project

Open the Makecode editor on your browser: https://makecode.microbit.org，click"**New Project**", Then you can **give a name** for your project.

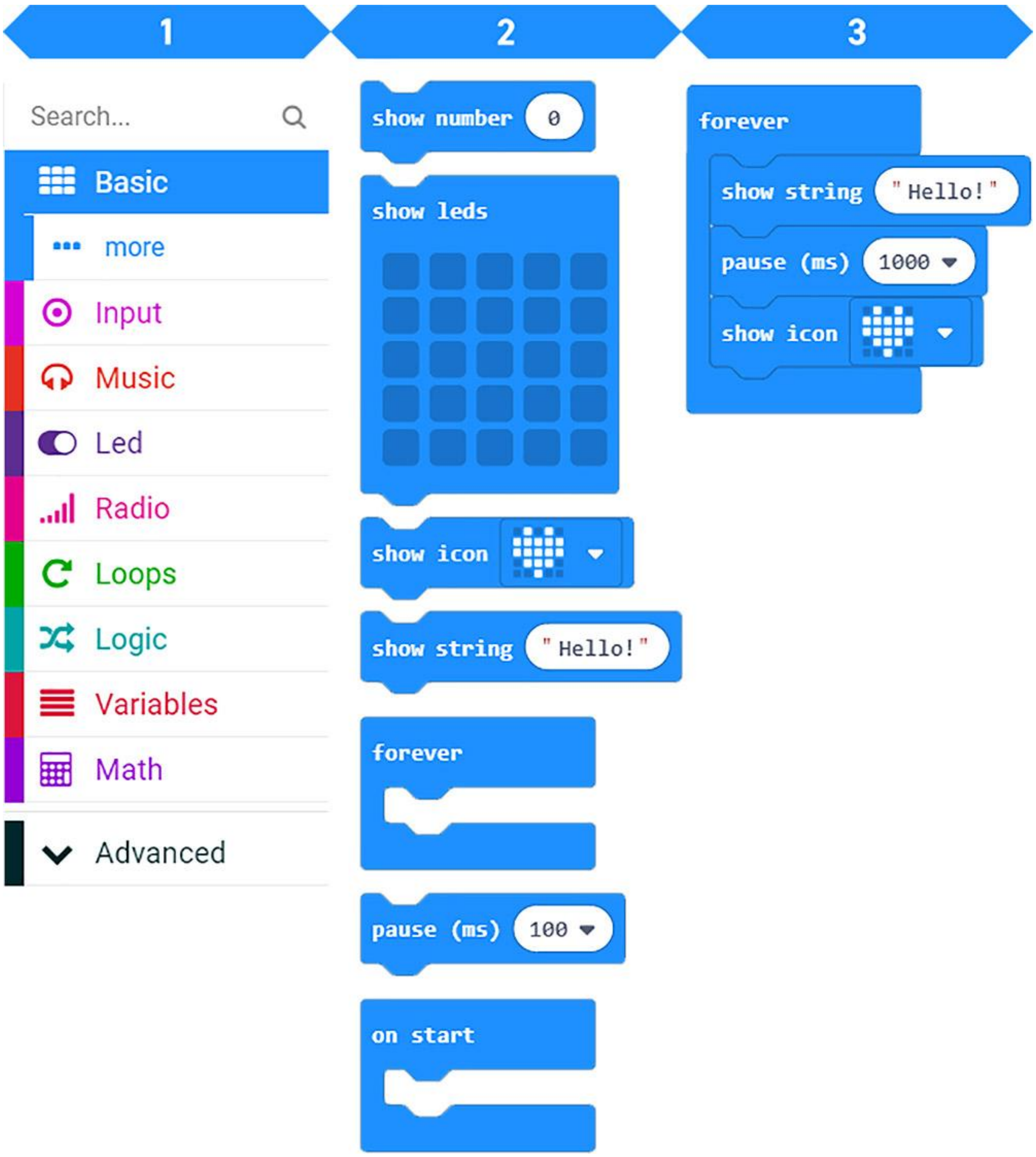


The newly created projects will be saved in the current browser. Just revisit the https://makecode.microbit.org website and find them in the project list.



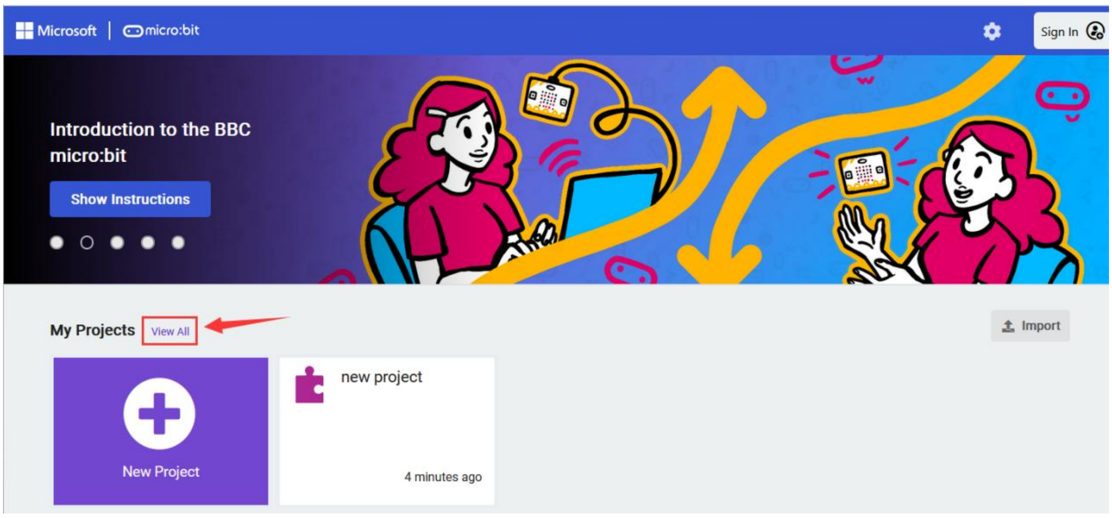
Selecting Blocks:

1. Select a block category from the list on the left-hand side of the page.
2. Select a block from the selected category, then drag it to the workspace area on the right.
3. Snap new blocks onto existing blocks in the workspace area. As the new blocks are dragged into the workspace, the editor highlights the connecting parts of each block when they are in a valid position to snap to existing blocks. Also, the shape of the blocks gives you an indication of where they might fit into your code blocks.

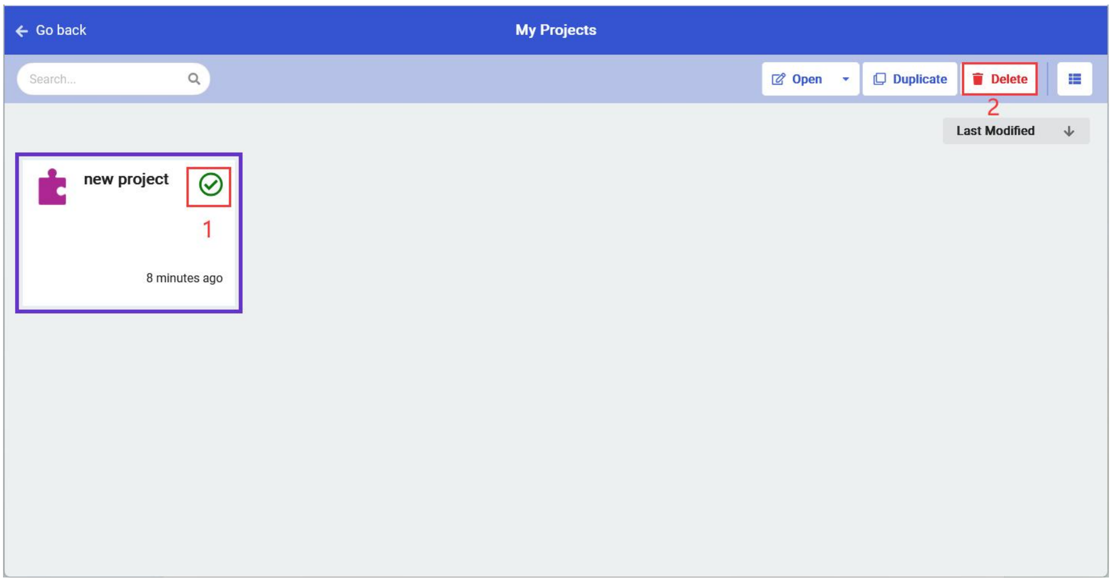


3.2 Delete a Project

Click“**View All**”:

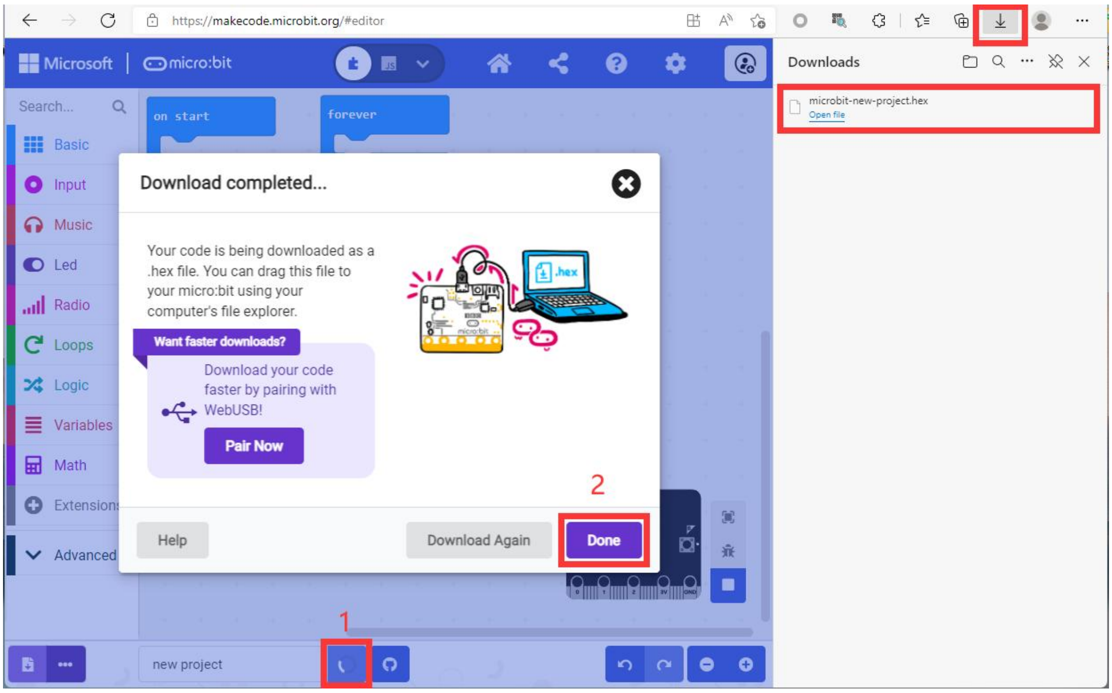


Select the project you want to delete, and click **Delete** button.



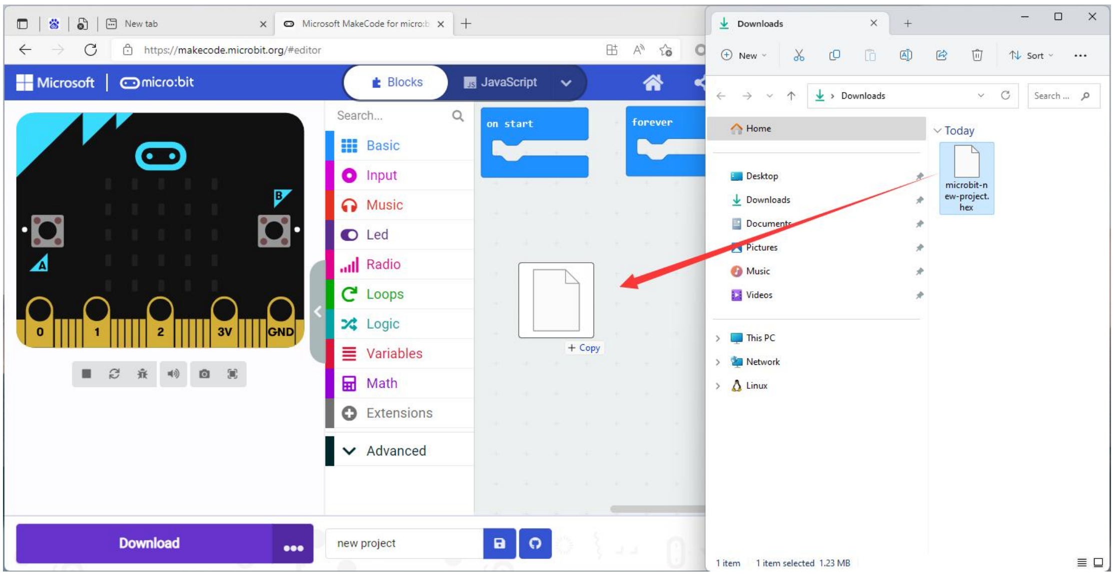
3.3 Save a project

Click the "**Save**" button, and then click the "**Done**" button to save the project to your computer, as shown below:



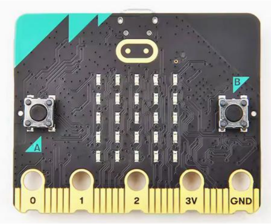
3.4 Import Files

Simply drag the local "HEX" project file to the work area of the MakeCode editor, as shown below:

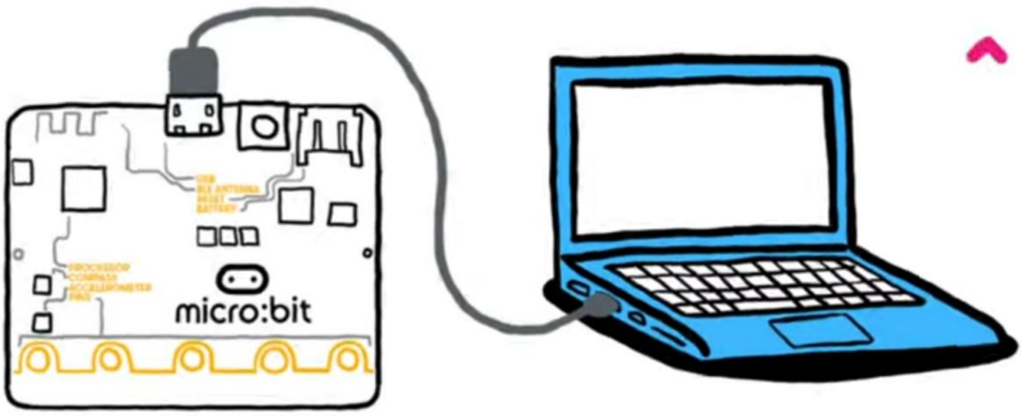


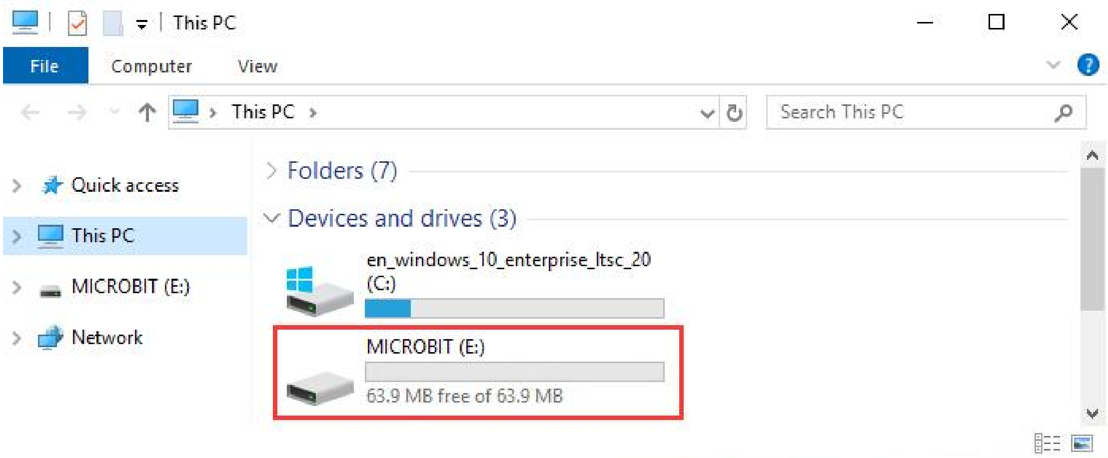
3.5 Upload code

Things you need:

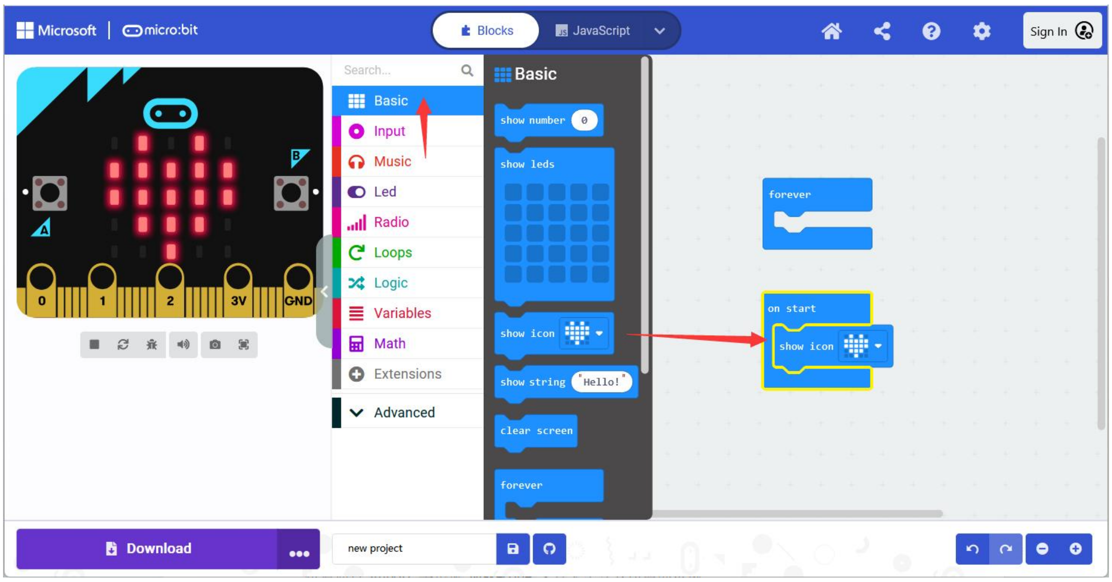
PC	Micro:bit v2.x.x	Micro USB Cable
		

Use a Micro USB cable to connect the micro:bit board to the PC. You will find a new USB disk called MICROBIT on the PC:

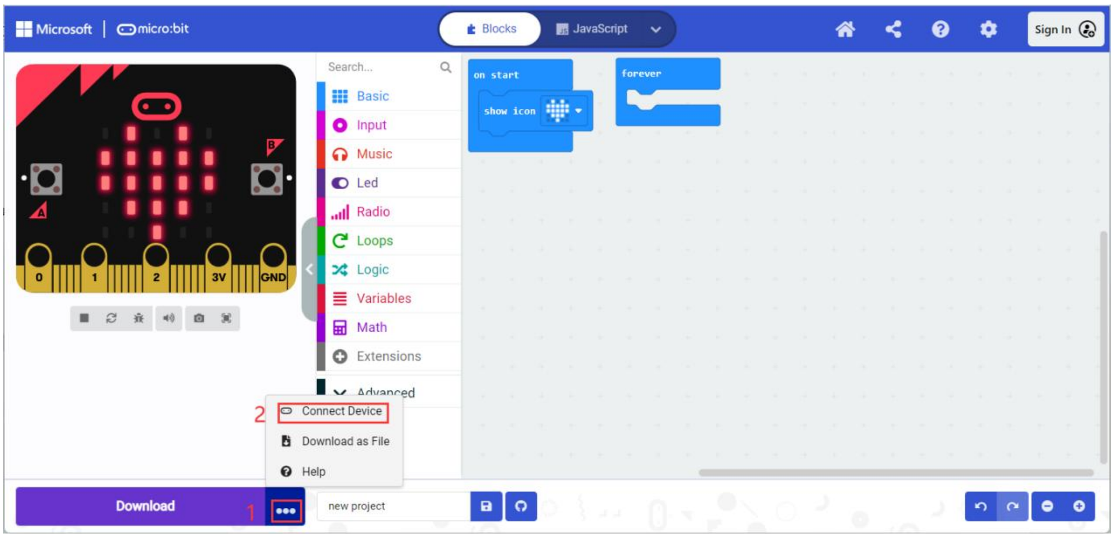


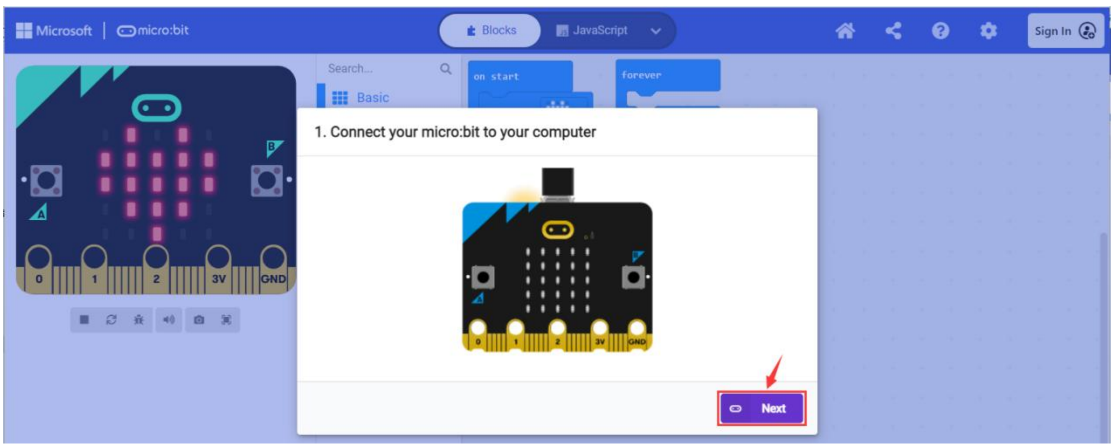


Open the MakeCode editor on your browser, hold down the left mouse button, and drag the **show icon** statement on the left to the working area on the right: https://makecode.microbit.org/#editor



Click on the **three dots** next to the Download button. Then click the "**Connect Device**", and then click the "**Next**" , as shown below:





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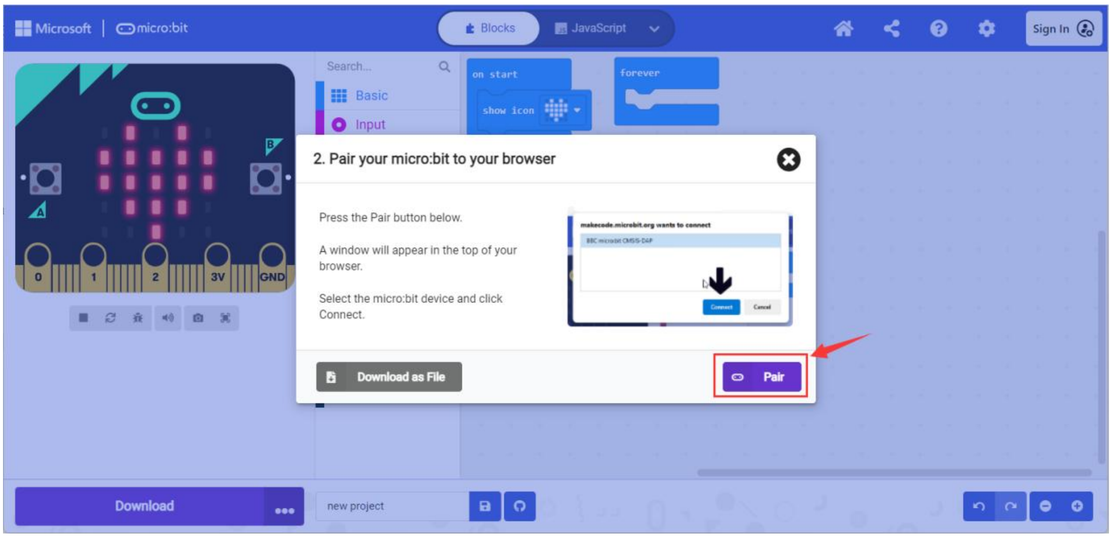
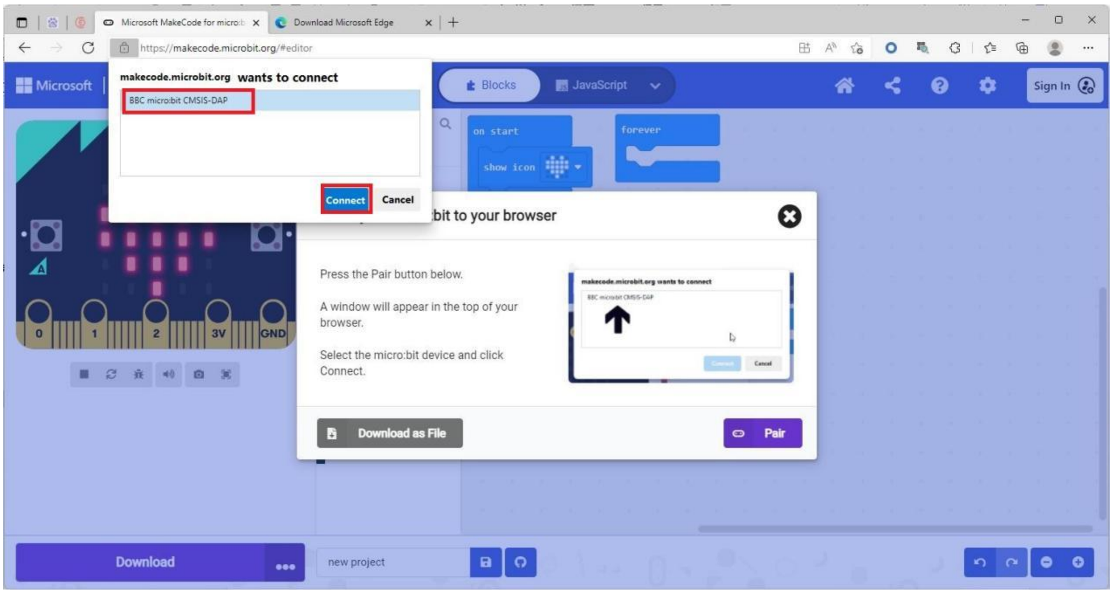
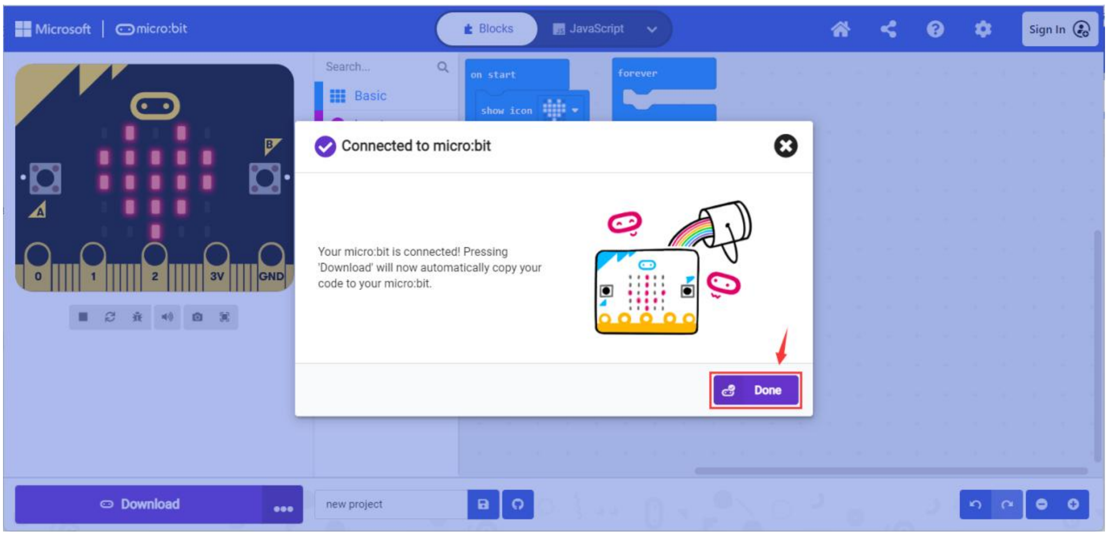
Click the "**Pair**" button:



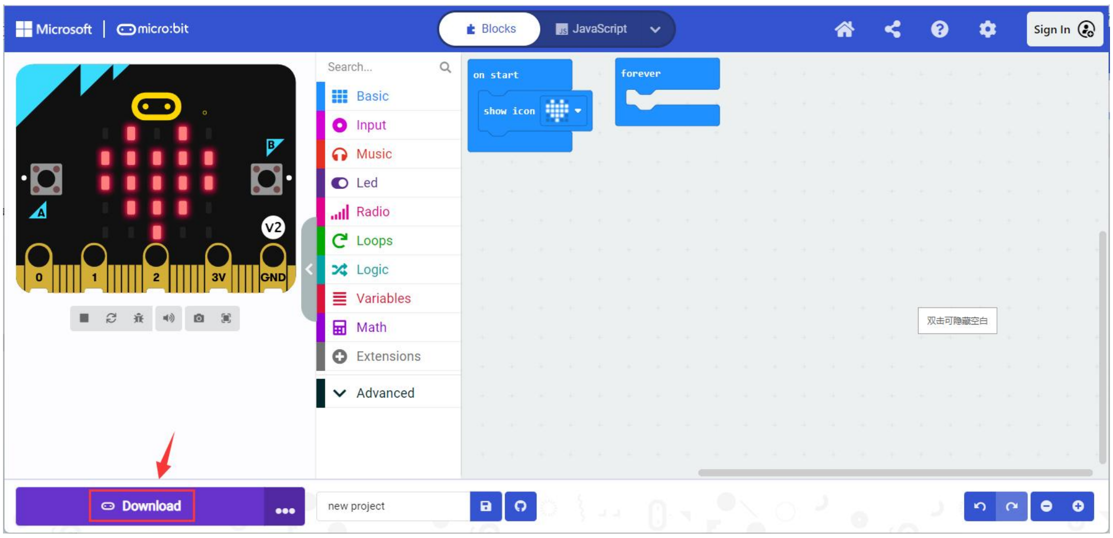
Select the Micro:bit board you want to connect to, and then click "**Connect**":

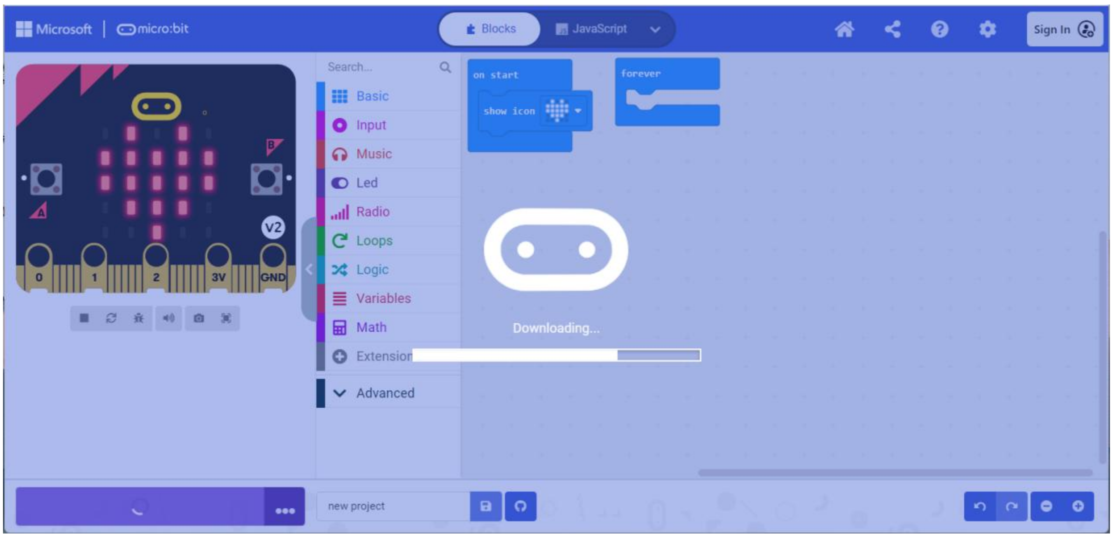


The connection is successful, click"**Done**":

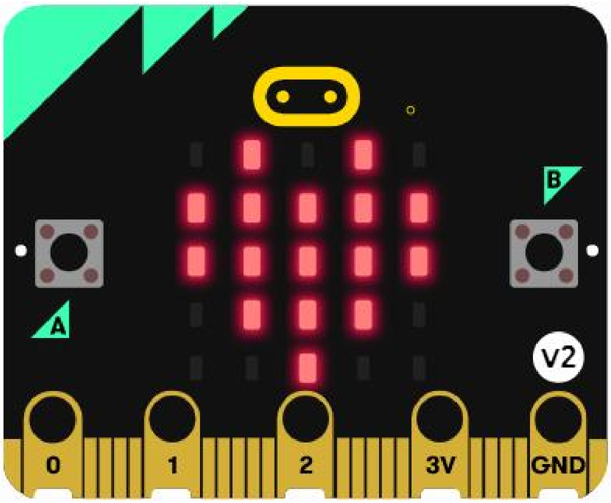


Click the "**Download**" button, you can flash the code to the Micro:bit with WebUSB.





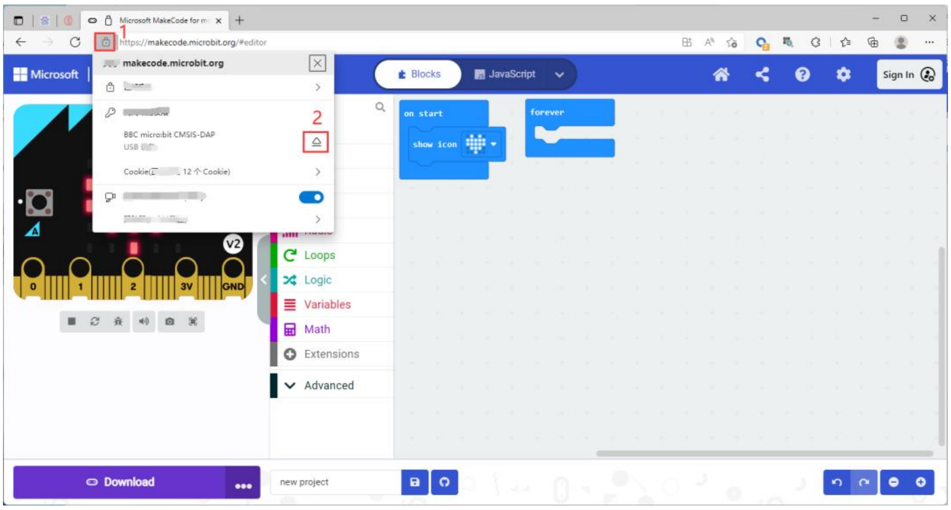
After uploading the code, the dot matrix of the Micro:bit board displays a heart shape:



3.7 Unpairing Micro:bit

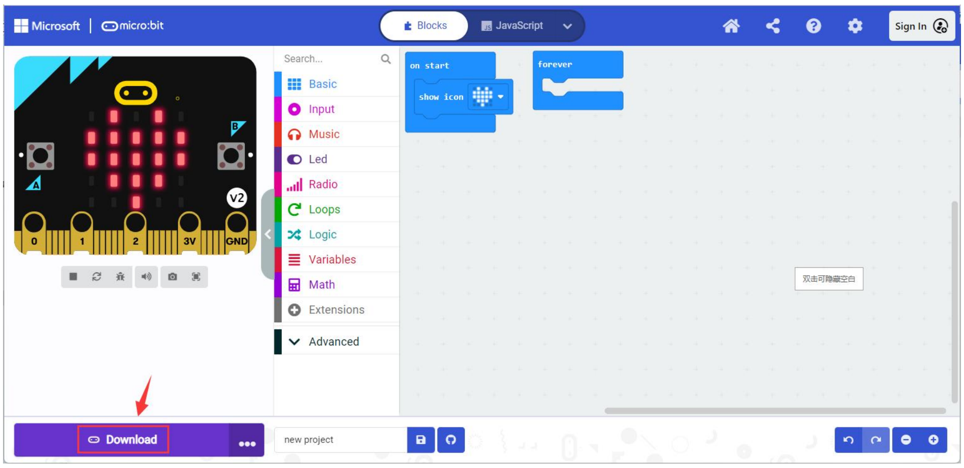
-Click the button to the left of your browser's search box.

-Select the Micro:bit device you want to disconnect and click the button to the right of it.

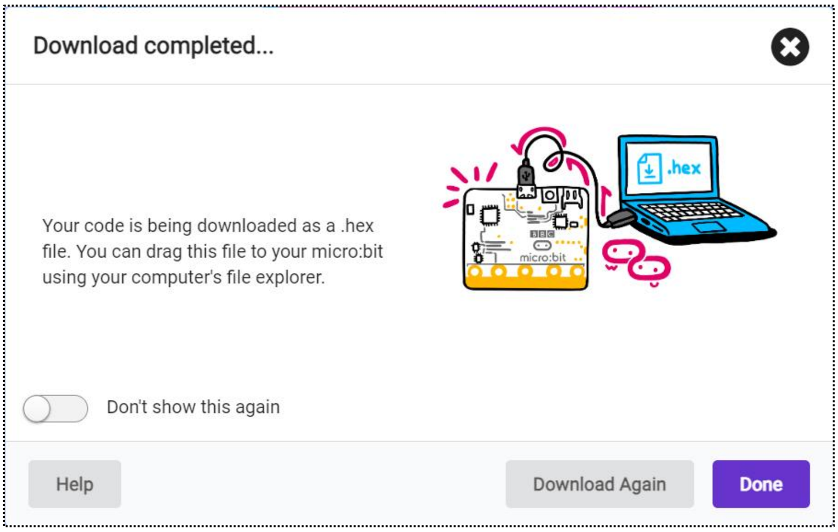


3.8 Upload the HEX file to the Micro:bit

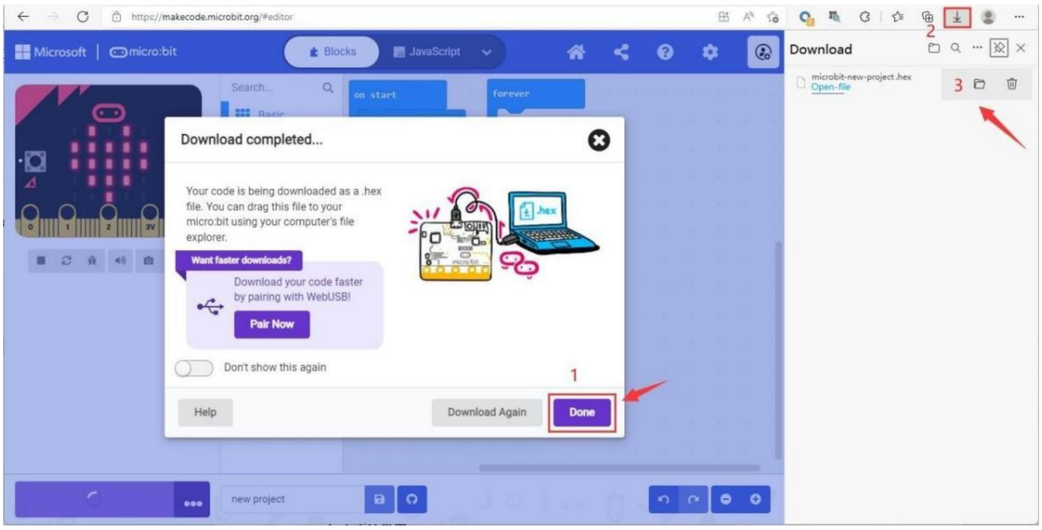
If the Micro:bit is not paired with Microsoft Edge or Google Chrome browser, or if you are using Safari/Firefox/Other browser that may not support WebUSB, directly click the "**Download**" button, the code won't transfer directly to your micro:bit, it will be downloaded as a .hexfile. Just like click the save icon to save a copy of the hex file to your computer.



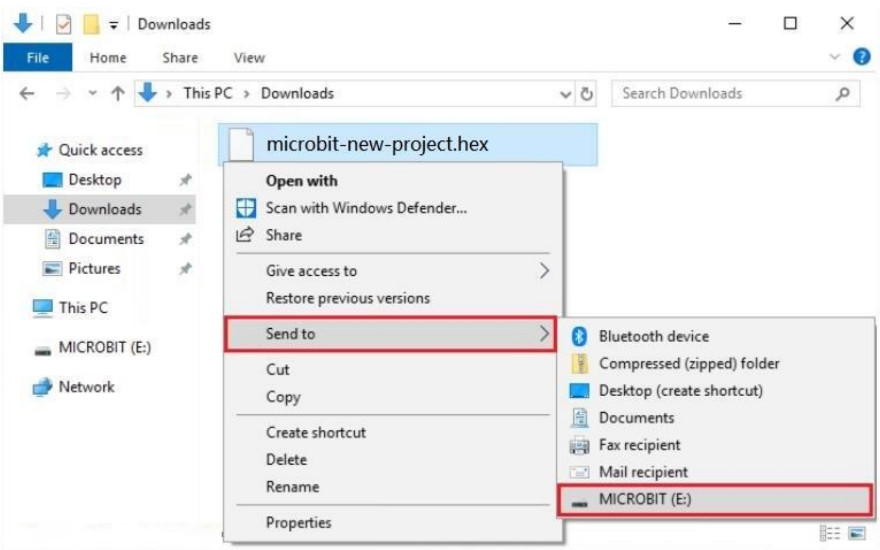
When the following interface appears, click the "⊗" button and click "**Done**":



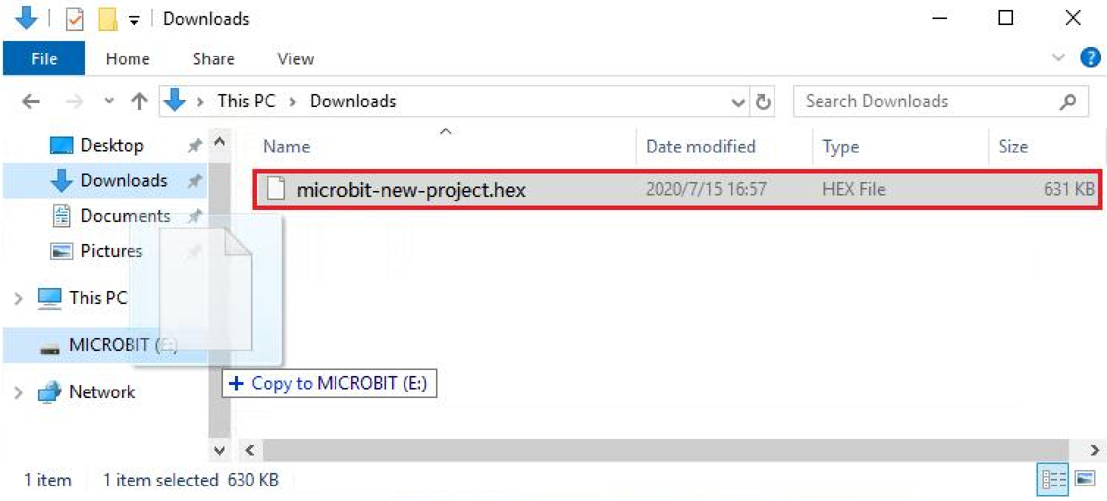
Find the downloaded hex file in the default save path of browser.



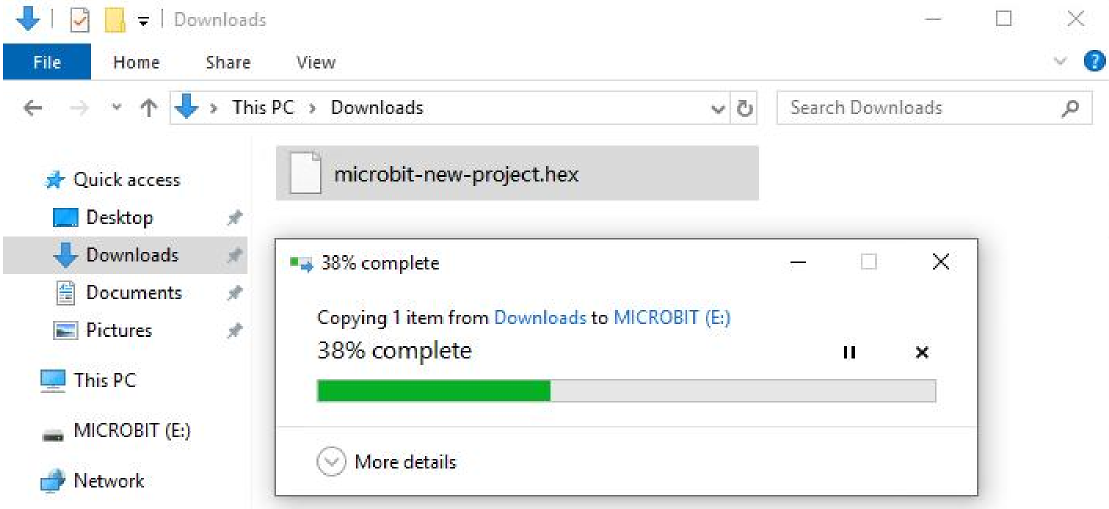
Then select the downloaded hex file，right click the mouse and click "Send to"，then you can send the hex file to your Micro:bit:



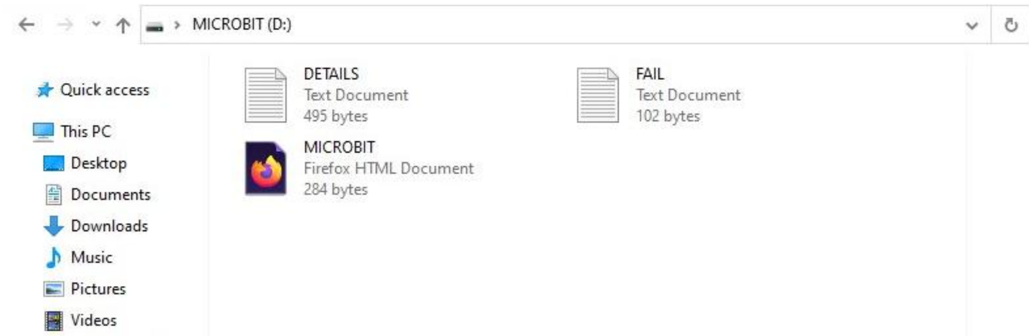
If you do not use the "**Send to**" button, you can directly drag the hex file to the Micro:bit:



The following interface indicates that the code is being uploaded, at the same time, the yellow LED on the back of the Micro:bit will also flash rapidly until the code upload is complete.



After the code upload is complete, the Micro:bit will disconnect and reconnect. If you look at the contents of the MICROBIT drive, you will not see the .hex file listed, this is normal, but your hex file will start automatically.



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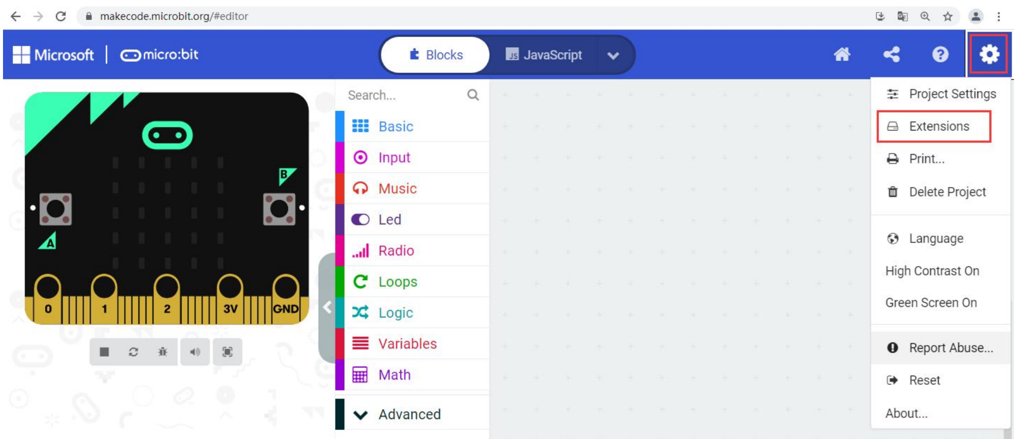
3.9 Learn the basic syntax of Makecode

The Micro:bit platform provides official MakeCode API and device usage documents for your reference.
To use APIs: https://makecode.microbit.org/reference
To use device: https://makecode.microbit.org/device
Logic and data types: https://makecode.microbit.org/blocks

4. Add Extension

4.1 Add Pybit extension

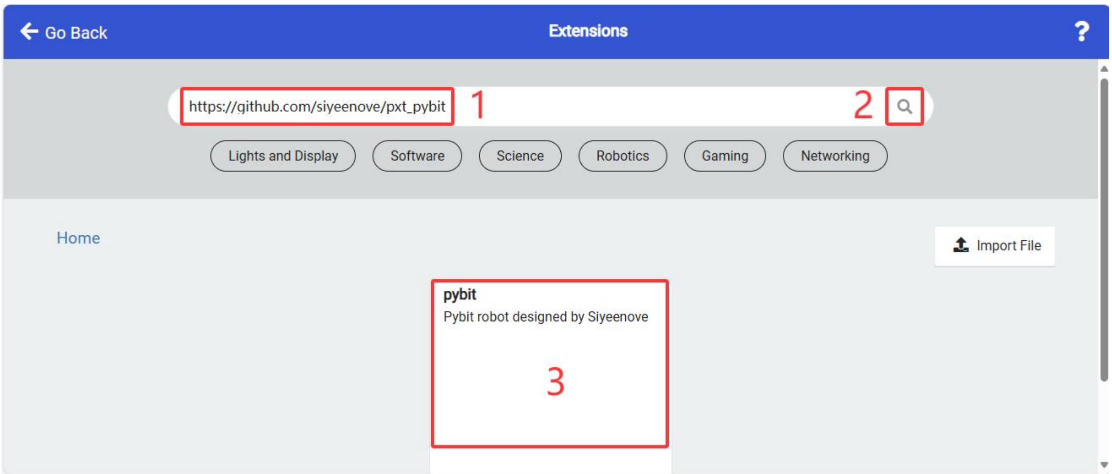
We have developed an extension for Pybit, which makes it easier for us to use MakeCode to program Pybit. The steps to add the extension to the MakeCode are as follows:
Click the **Settings** button in the upper right corner of the interface and click "**Extensions**"



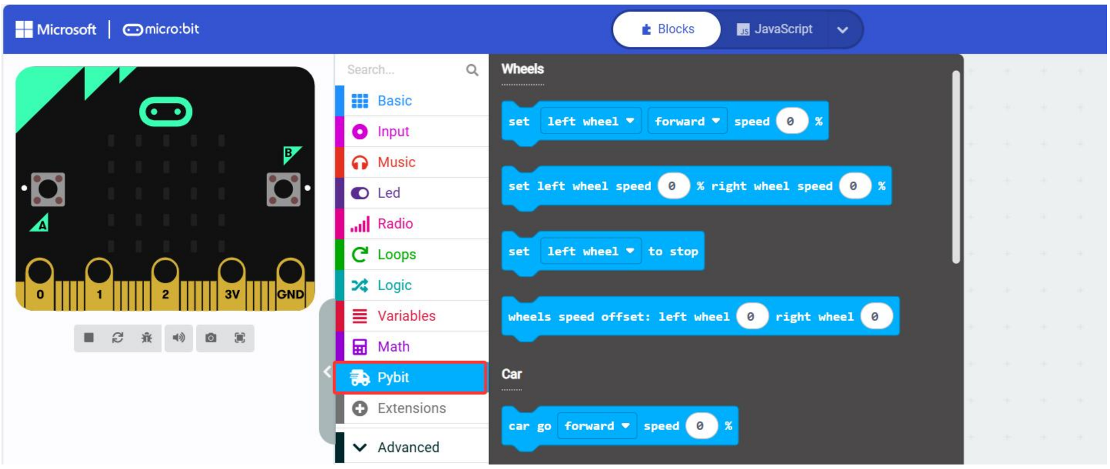
Extension link for Pybit：**https://github.com/siyeenove/pxt_pybit**

Copy the above link into the search box on the extension page and click the search button on the right.

Click on the extension named **pybit** in the search results.



After a few seconds the page will jump to the MakeCode main interface, and you will see the added **Pybit** extension in the toolbox list.



4.2 Add Neopixel extension

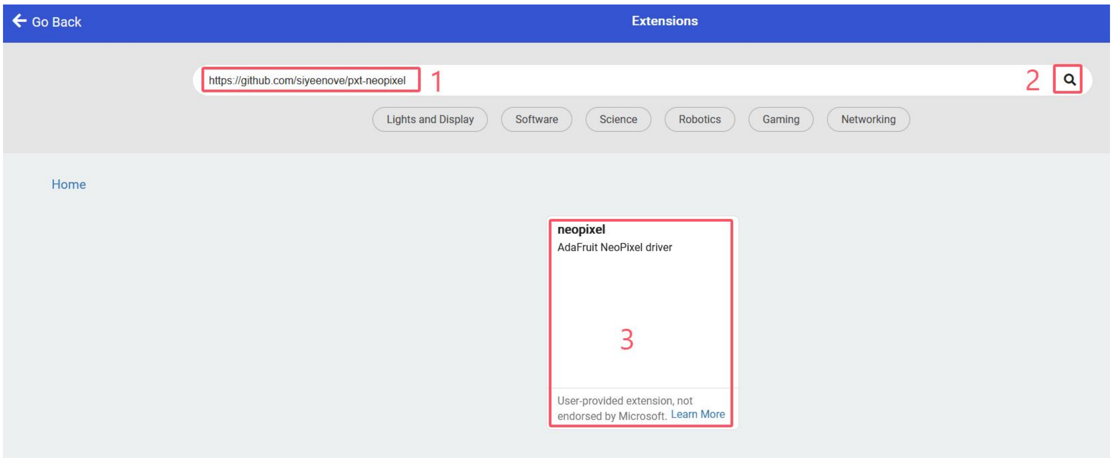
Since other Neopixel extensions lack Bluetooth compatibility, we developed a dedicated Neopixel extension for this robot's programming.

https://github.com/siyeenove/pxt-neopixel

Copy the above link into the search box on the extension page and click the search button on the right.

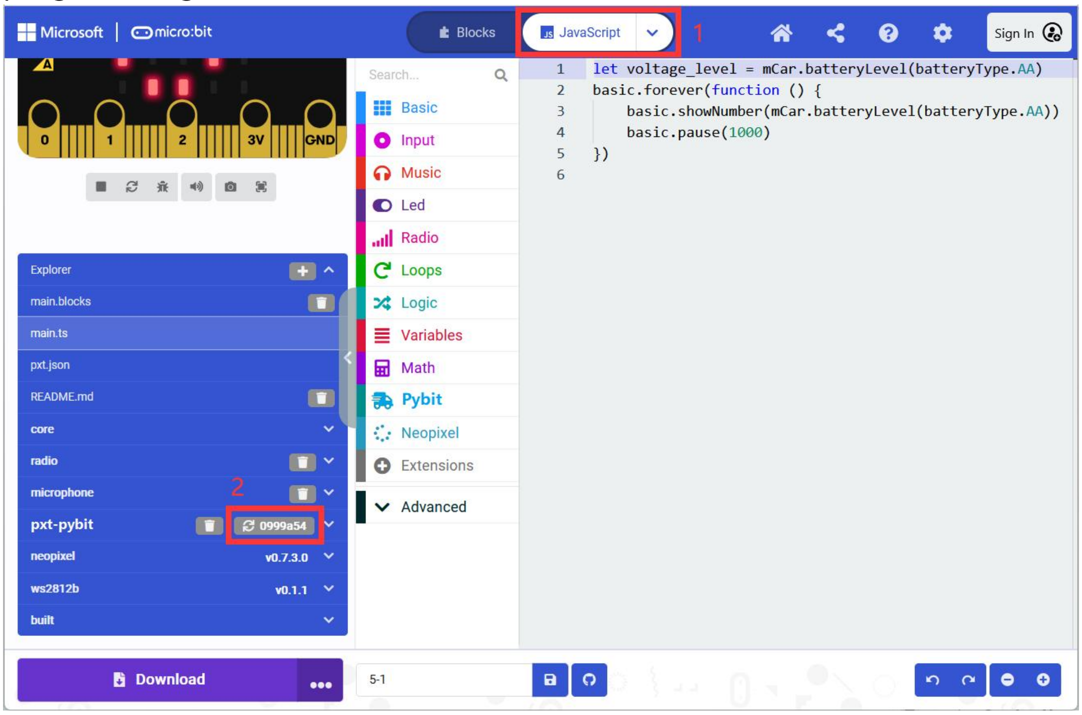
Click on the extension named **neopixel** in the search results.

After a few seconds the page will jump to the Makecode main interface, and you will see the added extension in the toolbox list.



4.3 Refresh pybit extension

Open the project with the **pybit** extension added, and switch to the JavaScript programming interface to refresh the extension:



After refreshing, switch back to the "Blocks" interface.

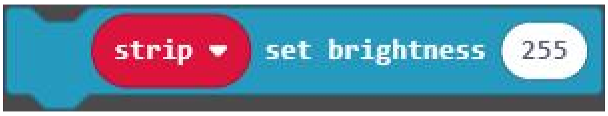
4.4 Parsing of pybit extension statement

All MakeCode statements based on pybit are integrated in the Pybit extension package. The statement analysis is as follows:

Wheels	`set left wheel ▾ forward ▾ speed 0 %` Set the Pybit's wheels to move forward or backward, and at what speed.
	`set left wheel speed 0 % right wheel speed 0 %` Set the speed and direction of the left and right wheels of the Pybit.
	`set left wheel ▾ to stop` Set the Pybit wheels to stop turning.
	`wheels speed offset: left wheel 0 right wheel 0` Adjust the speed of the left and right wheels of Pybit and save them permanently inside Pybit. This statement can be used to adjust the speed of Pybit when it cannot go in a straight line.
Car	`car go forward ▾ speed 0 %` Control the speed at which Pybit moves forward or backward.
	`car turn left ▾ turn rate 0 % speed 0 %` Controls the turning rate and speed of the Pybit to turn left or right. The greater the turning rate, the smaller the turning angle.
	`car turn left ▾ at place speed 0 %` Control the speed at which Pybit turns in place.
	`car stop` The Pybit wheels stopped turning.
RGB LED headlights	`set left headlight ▾ color:` Colors from the Pybit's headlight color palette.

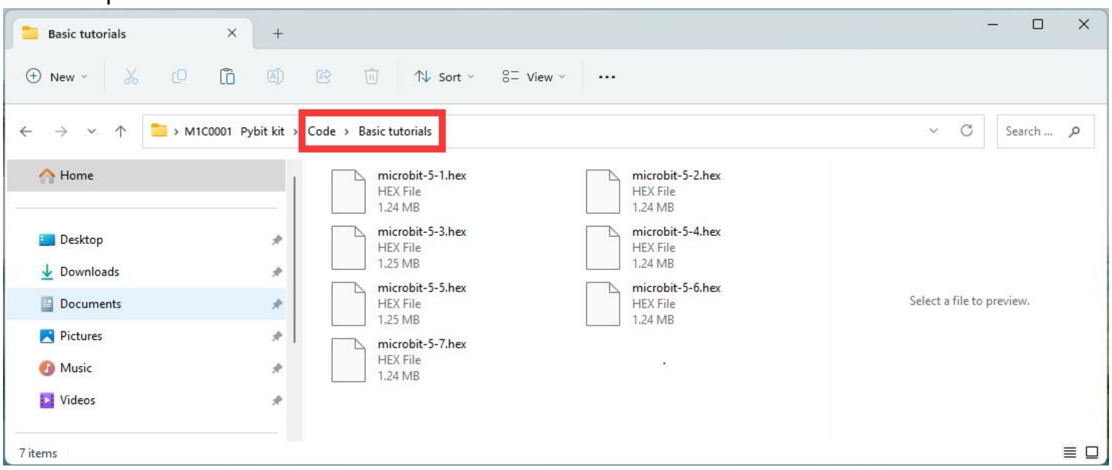
RGB LED headlights	`set left headlight ▾ color: red 0 green 0 blue 0` By changing the three color values, the Pybit's headlights can be lit in different colors.
	`turn off all RGB LED headlights` Turn off the Pybit's headlights.
Tracking sensor	`read tracking sensor status` Read the values detected by the Pybit 3 line tracking module and save them in internal variables.
	`tracking sensor state is ○ ○ ○ ▾` Determine whether the value of the 3-way line tracking sensor is equal to the value in the statement tab. Before using this statement, you need to call the "Get the status value of the tracking sensor" statement once.
	`tracking sensor value` The value of the 3-way line tracking sensor in the variable. Before using this statement, you need to call the "Get the status value of the tracking sensor" statement once.
	`left ▾ tracking sensor state is ● ▾` Determine whether the value of one of the three line tracking sensors is equal to the value in the statement tab. Before using this statement, you need to call the "Get the status value of the tracking sensor" statement once.
Sonar sensor	`sonar distance cm ▾` Read the distance value of the ultrasonic module.
Infrared sensor	`on IR receiving` Infrared receiving loop function, this function has been processing the received infrared data.

Infrared sensor	`IR button 1 ▾ is pressed` Determine whether the data received by the infrared receiver is equal to the key value in the Statement tab.
	`IR value` Read the data received by the infrared receiver. Before using this statement, use the "if" statement to determine when the return value is not 0, and then the data will be valid.
Expansion port	`set S1 ▾ 90° ▾ servo angle to 0 °` to drive the servo statement, which can drive 90, 180, and 270 degree servos; the interface is the S1, S2, and S3 expansion port on the Pybit.
	`set S1 ▾ 360° servo speed to 0 %` The driving statement of 360-degree servo; the interface is S1, S2, S3 expansion port on Pybit.
Baterry	`battery level: 3 AA batteries ▾` Statement to sets the battery model and reads the battery voltage, returning the battery level between 0 and 100%. **It is recommended to execute this statement once when the machine is turned on.**
APP command	`left ▾ textview display " "` Send the string to the left or right text display box of SIYEENOVE APP.
	`" " equal F-pressed ▾` Determine whether the data received by the Bluetooth of the Micro:bit is the key value of SIYEENOVE APP.
Other	`version number` Read the version number of Pybit and return a string value.
Neopixel	`set strip ▾ to NeoPixel at pin P8 ▾ with 2 leds as RGB (GRB format) ▾` Use the P8 pin of micro:bit to drive the two WS2812 RGB lights at the bottom of Pybit.
	`strip ▾ show color red ▾` The two WS2812 RGB lights on the bottom of the Pybit light up the

Neopixel	colors in the tabs.
	 Set the brightness of the two WS2812 RGB lights at the bottom of the Pybit.
	 Turn off all WS2812 RGB lights.
	......

5. Basic Tutorials

The sample codes for the basic tutorials are all saved in the "Code -> Basic tutorials" file:



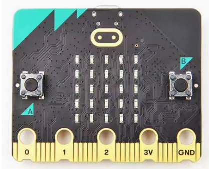
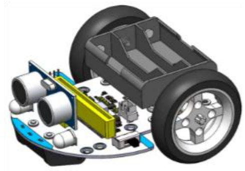
Reminder!!!
You don't need to add the Pybit extension if you import and use the code we provided. ---> Recommend!
If you create a new project and drag and drop blocks to build code, you need to manually add the Pybit extension, please refer to sections 3.1 and 4.1.

5.1 Battery Fuel Gauge

Goal

Set the battery type and read the battery voltage level.

Things you need：

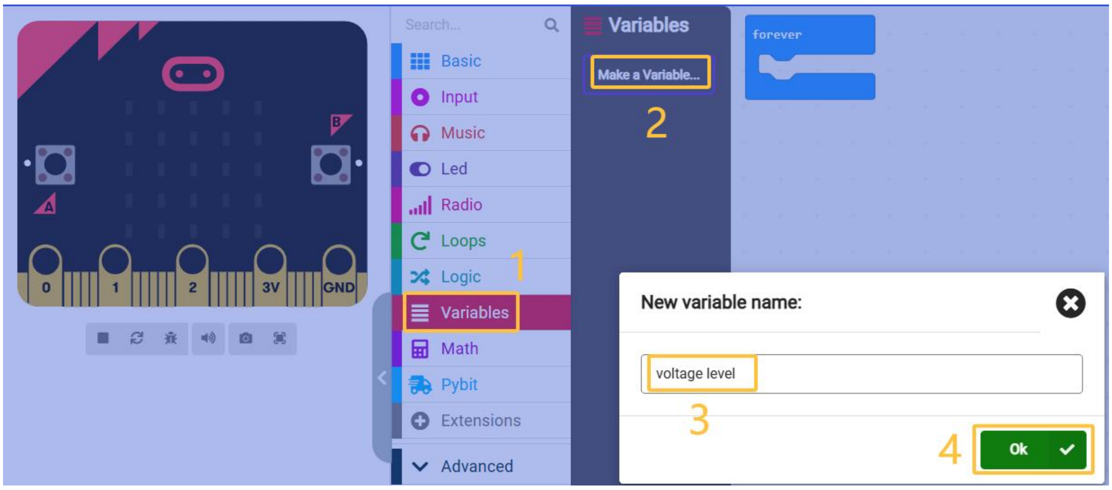
Computer	Micro:bit v2.x.x	Micro USB Cable	Pybit
			

Programming

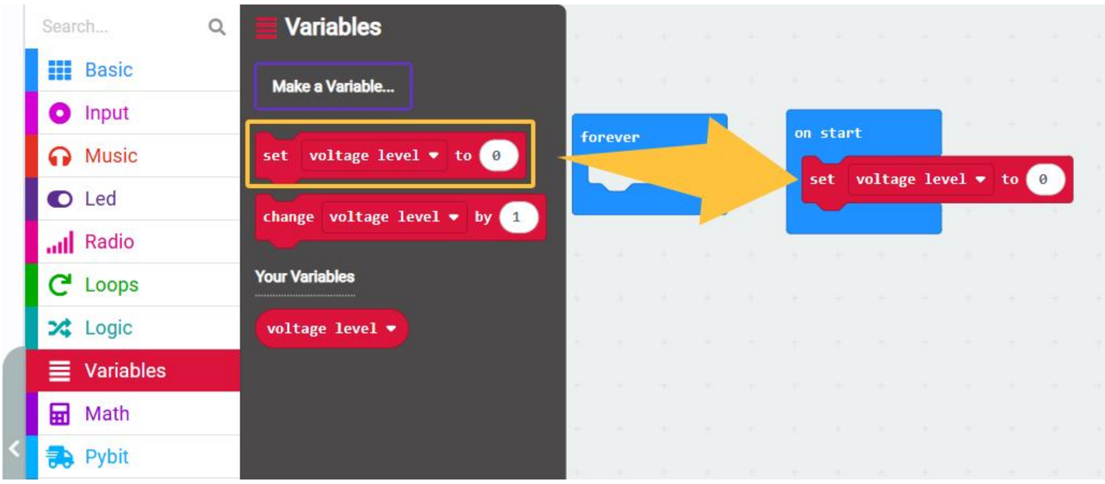
Import the hex file code we provided into the MakeCode editor and download it to Micro:bit.

If drag to build a new code, you need to add the Pybit extension.

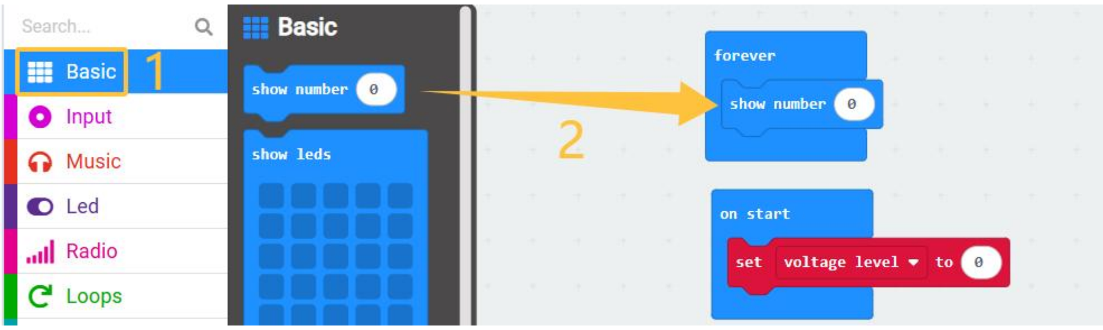
1.Click "Variables" > "Make a Variables" > New variable name: "voltage level".



2.Drag "set voltage level to 0" and snap it into "on start".

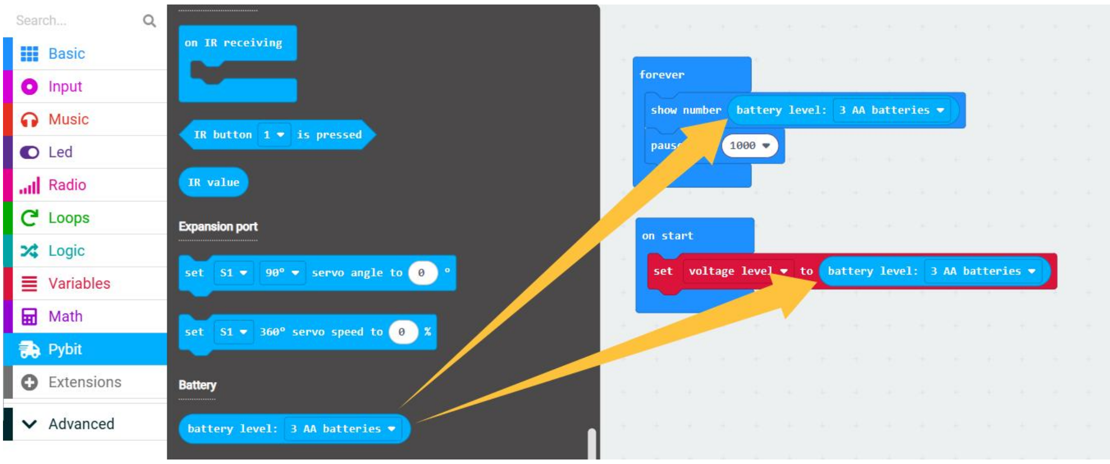


3. Click "Basic", drag "show number" and snap it into "forever".



4.Click "Basic", drag "show number" and snap it into "forever", then change it to pause 1000ms.

5.Click "Pybit", snap "battery: 3 AA batteries" between "set voltage level to 0" and "show number".



Code Summary:

• Create a variable.

• Set the battery type in the "on start" block and read the battery voltage level;

• Display the battery voltage level in the "forever" block.

Consequence

The dot matrix of Micro:bot displays the voltage level, with a value range of 0--100%. When the battery voltage level is less than or equal to 20%, the battery indicator light is always on; otherwise, it flashes.

Note

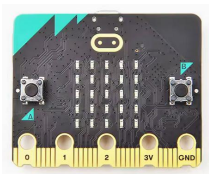
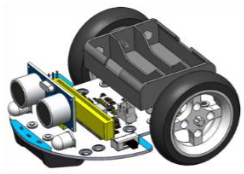
The statement to read the battery voltage level must be executed once in the program to set the battery type, otherwise the battery voltage indicator will not work properly!

5.2 Headlights

Goal

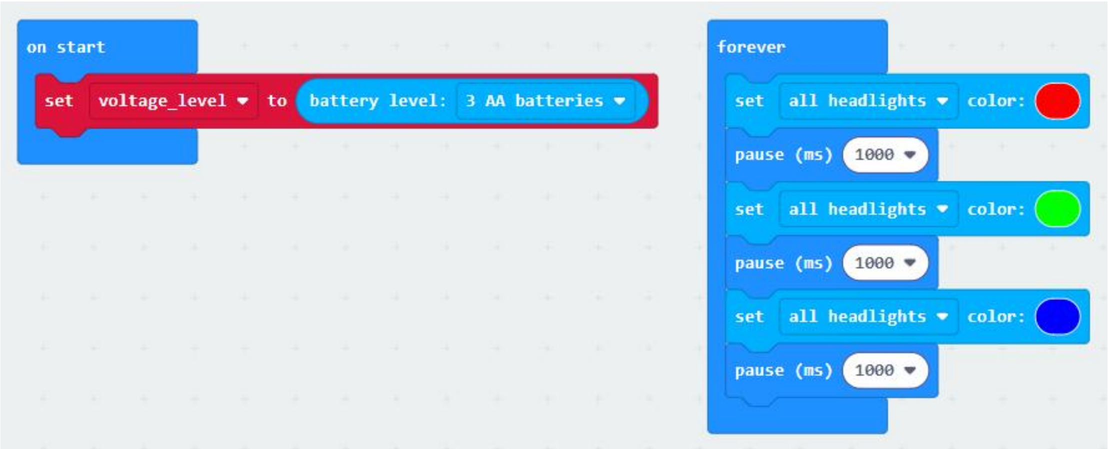
Make the Pybit's headlights flash.

Things you need:

Computer	Micro:bit v2.x.x	Micro USB cable	Pybit
			

Programming

Directly import the hex file code we provided into the Makecode editor and download it to Micro:bit. If drag to build a new code, remember to add the Pybit extension.



• Set the battery type in the "on start" block.

• Light up both headlights in the "forever" block.

Consequence

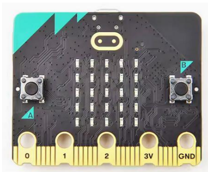
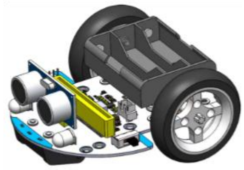
The two headlights of the Pybit light up in a cycle: red -> delay 1 second -> green -> delay 1 second -> blue -> delay 1 second.

5.3 WS2812 RGB Ambient Lights

Goal

Make the two WS2812 RGB lights on the bottom of the Pybit light up.

Things you need：

Computer	Micro:bit v2.x.x	Micro USB cable	Pybit
			

Programming

Directly import the hex file code we provided into the Makecode editor and download it to Micro:bit. If drag to build a new code, remember to add the Pybit extension.

• Set the battery type in the "on start" block.

• Light up the two WS2812 RGB ambient lights at the bottom of the Pybit on the "forever" block.

Consequence

The two WS2812 RGB ambient lights at the bottom of the Pybit light up in a cycle: red -> delay 1 second -> green -> delay 1 second -> blue -> delay 1 second.

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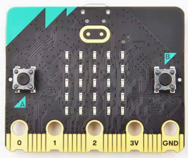
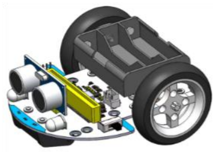
5.4 Infrared Receiver

Goal

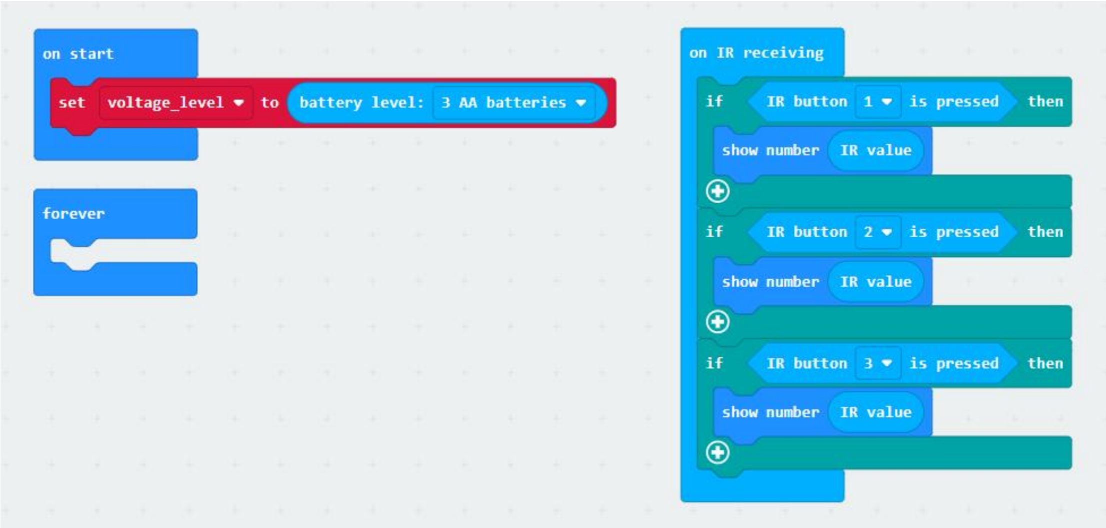
Read the key values of the infrared remote controller through the infrared receiver of Pybit.

Things you need：

Computer	Micro:bit v2.x.x	Micro USB cable	Pybit	IR Remote
				

Programming

Directly import the hex file code we provided into the Makecode editor and download it to Micro:bit. If drag to build a new code, remember to add the Pybit extension.



• Set the battery type in the "on start" block.

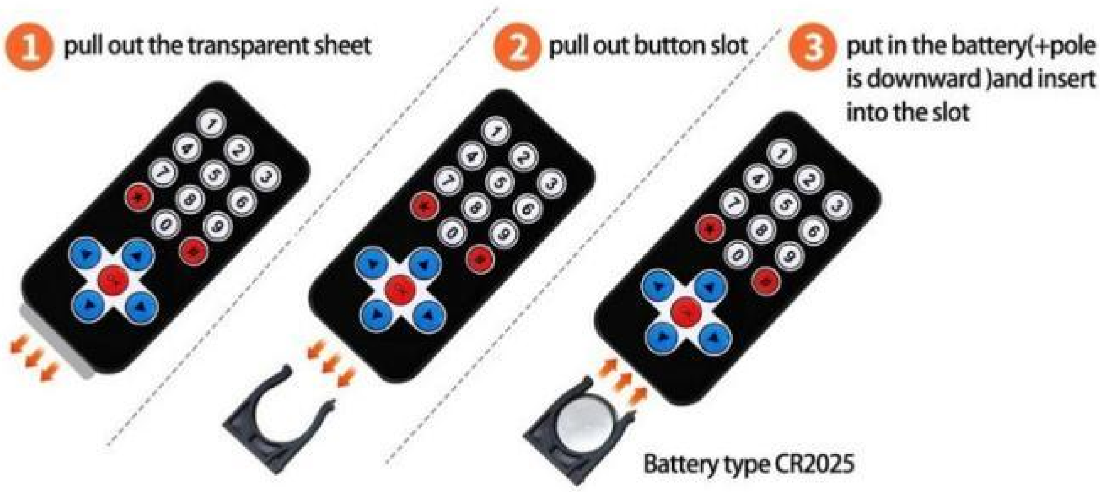
• On the "forever" block, use 3 judgment statements in the infrared data receiving block to check whether the 1, 2 or 3 button is pressed. If a button is pressed, display the button value.

Consequence

Use the infrared remote controller to press the button toward the infrared receiver on the Pybit. The dot matrix on the Micro:bit will display the infrared remote controller button value.

Note：

When using the infrared remote control, the plastic battery cover at the bottom must be removed and the batteries must be installed correctly.

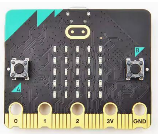
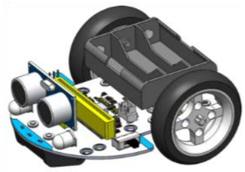
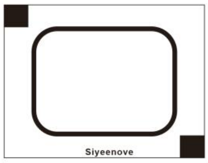


5.5 Infrared tube sensor (line tracking sensor)

Goal

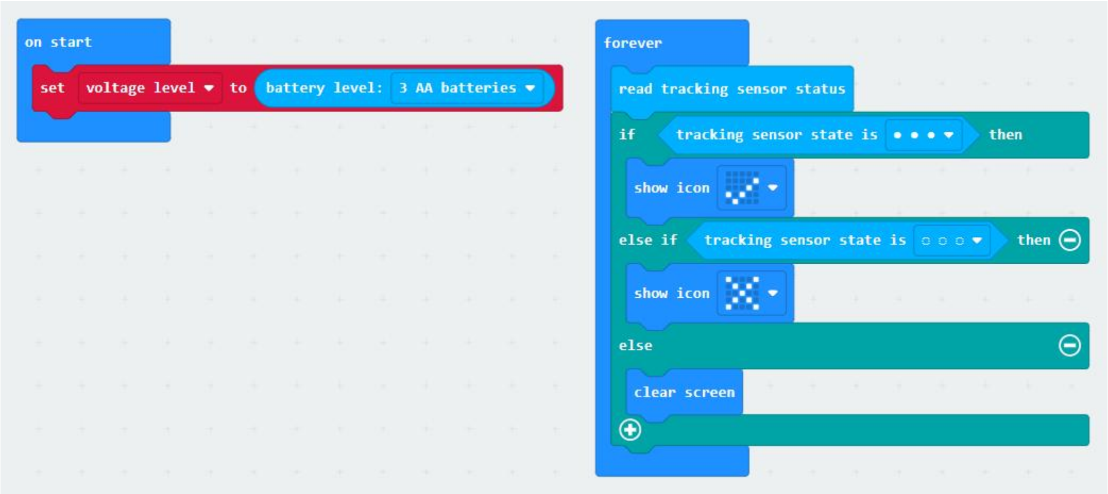
Let Pybit's three infrared black and white line sensors identify black and white area.

Things you need：

Computer	Micro:bit v2.x.x	Micro USB cable	Pybit	Map
				

Programming

Directly import the hex file code we provided into the Makecode editor and download it to Micro:bit. If drag to build a new code, remember to add the Pybit extension.



• Set the battery type in the "on start" block.

• Read the values of the three sensors in the "forever" block, and then use three judgment statements to process whether a black line is detected. If all three sensors detect black, "√" is displayed, and if none of them detect black, "×" is displayed.

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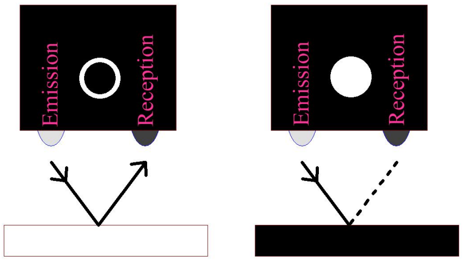
Consequence

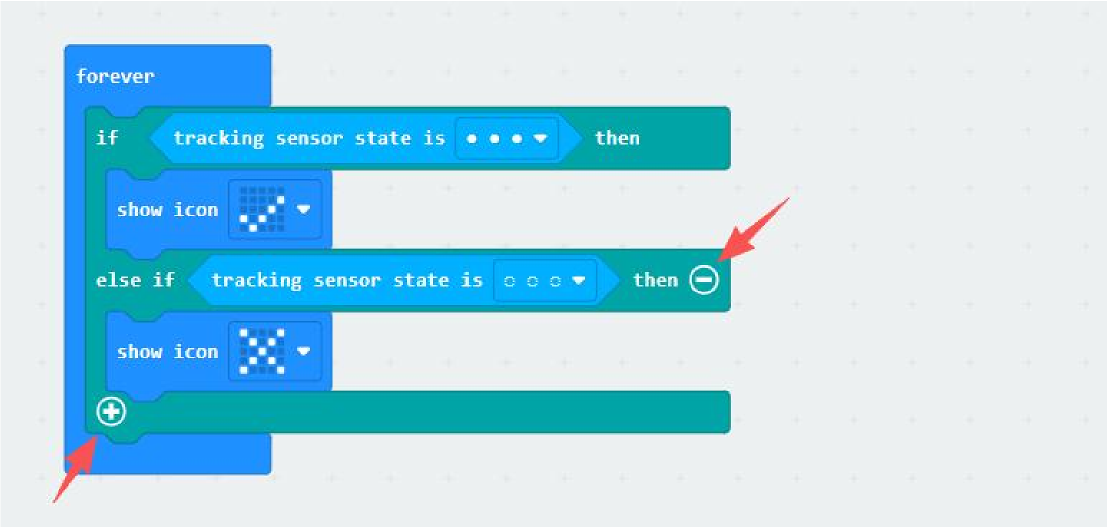
Black objects with rough surfaces have a good absorption effect on infrared rays. When the infrared rays emitted by the sensor hit the black line, the infrared rays are absorbed; when it hits the white object, the infrared signal is reflected back.



After uploading the code, put all three infrared sensors of Pybit in the black area of the map, and the Micro:bit dot matrix will display "√"; if all three infrared sensors are placed in the white area of the map, the Micro:bit dot matrix will display "×";

Note

By clicking the "⊕" or "⊖" button, you can switch to an "if...", "if .. else if..." or "if ... else if ... else" statement:



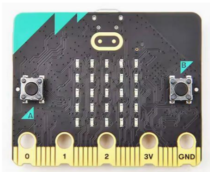
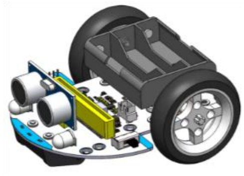
If placed under intense sunlight, the infrared rays in the sunlight can significantly interfere with the infrared line tracking sensor, causing abnormal operation of the Micro:bit car's infrared line tracking sensor.

5.6 Ultrasonic Sensor

Goal

The ultrasonic module measures the distance to obstacles.

Things you need:

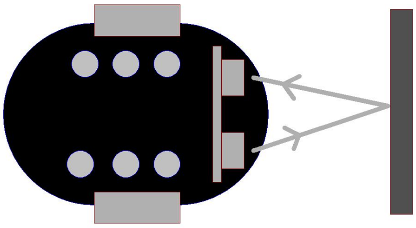
Computer	Micro:bit v2.x.x	Micro USB cable	Pybit
			

Programming

• Set the battery type in the "on start" block.

• Read the distance measured by the ultrasonic module in the "forever" block and display it in the dot matrix.

Consequence



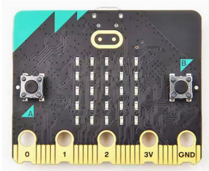
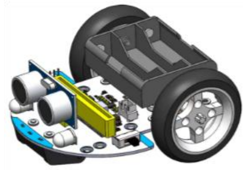
The dot matrix of Micro:bot displays the measured obstacle distance in centimeters.

5.7 Drive wheels

Goal

Learn how to drive the Pybit's 2 motors

Things you need：

Computer	Micro:bit v2.x.x	Micro USB cable	Pybit
			

Programming

Directly import the hex file code we provided into the Makecode editor and download it to Micro:bit. If drag to build a new code, remember to add the Pybit extension.

• Set the battery type in the "on start" block.

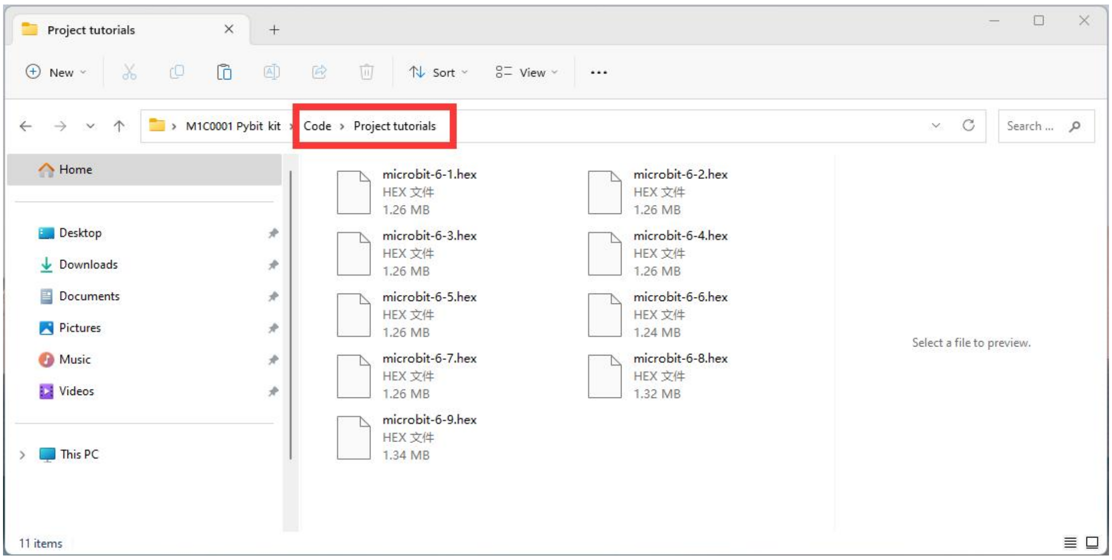
• In the "forever" block, let the two motors rotate in opposite directions for one second and stop for one second.

Consequence

Pybit rotates left and right on the spot.

6. Play with Pybit

The sample codes for the project tutorials are all saved in the "Code -> Project tutorials" file:



Reminder!!!

You don't need to add the Pybit extension if you import and use the code we provided. ---> Recommend!

If you create a new project and drag and drop blocks to build code, you need to manually add the Pybit extension, please refer to sections 3.1 and 4.1.

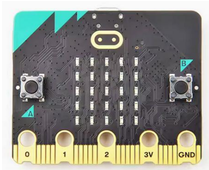
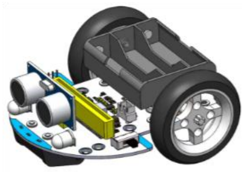
6.1 Evenly accelerate and go straight

Goal

If the motor speed of Pybit is very fast, the front universal wheel will leave the ground at the moment of starting.

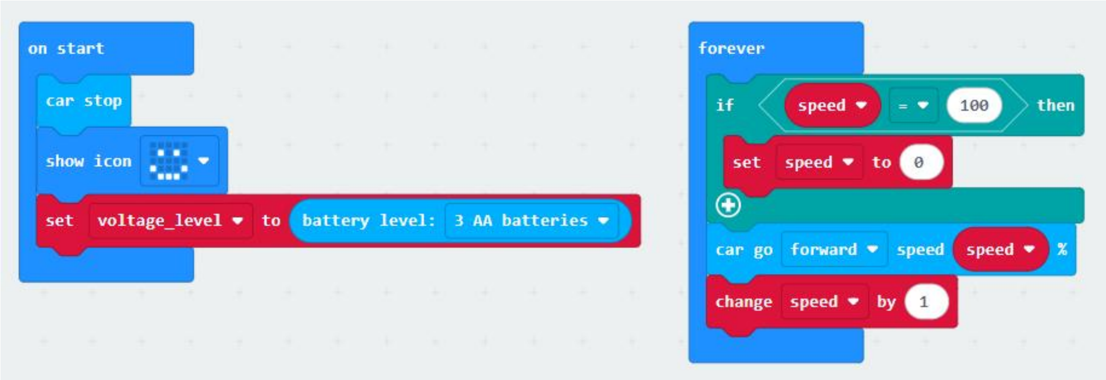
In this case, we let Pybit accelerate evenly and complete a smooth start.

Things you need:

Computer	Micro:bit v2.x.x	Micro USB cable	Pybit
			

Programming

Directly import the hex file code we provided into the Makecode editor and download it to Micro:bit. If drag to build a new code, remember to add the Pybit extension.



• In the "on start" block, stop the car, add a smiley face icon, and then set the battery type;

• Insert the car forward speed block in the "forever" block, set the speed value to speed, and then increase the speed by 1.

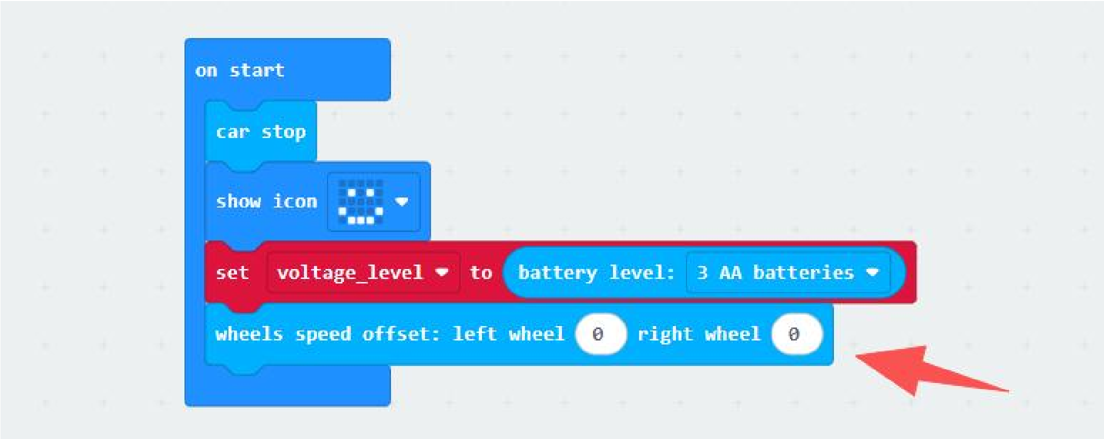
• And if the speed is equal to 100, the speed is already the maximum value, set the speed to 0, and start again.

Consequence

The car starts at a constant speed, stops at a certain speed and then starts again at a constant speed, so that the front wheels will not leave the ground due to excessive speed.

Note

If the Pybit cannot go straight, you can add a motor speed control statement in the "on start" block. The parameters of this statement will be permanently saved in the Pybit.



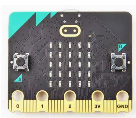
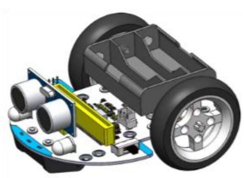
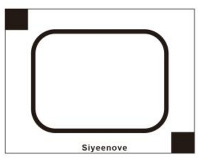
This language block adjust the speed of the left and right wheels of Pybit and save them permanently inside Pybit. This statement can be used to adjust the speed of Pybit when it cannot go in a straight line.

6.2 Follow the black line

Goal

Let Pybit run in circles along the black line.

Things you need：

Computer	[Micro:bit v2.x.x]()	Micro USB cable	Pybit	Map
				

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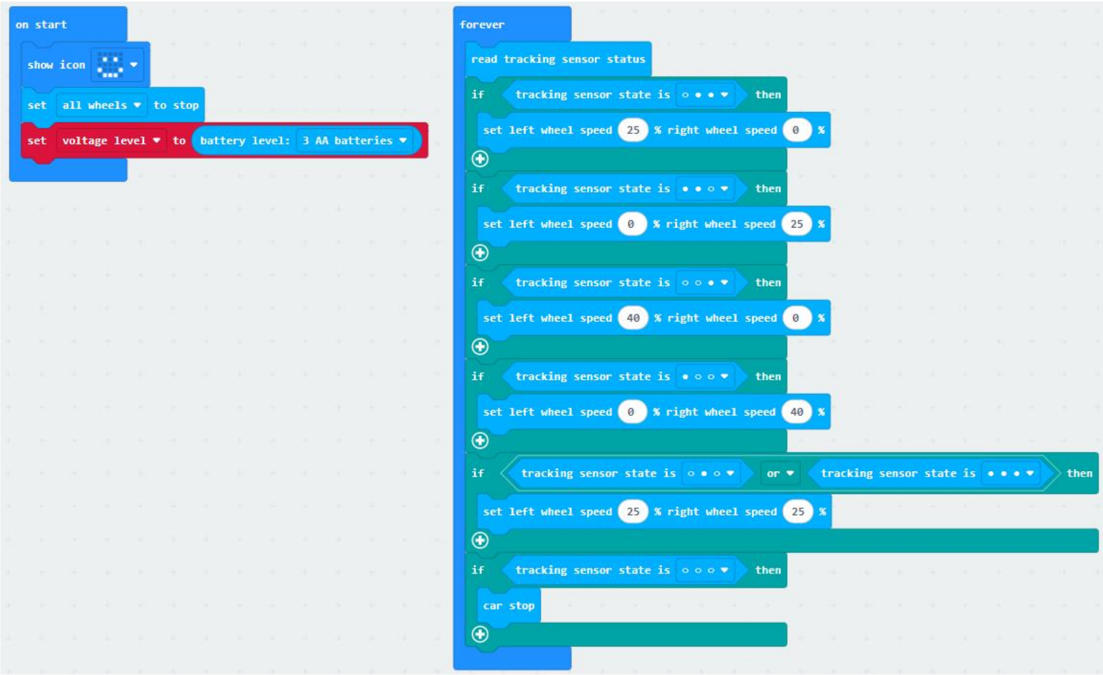
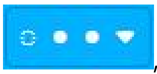
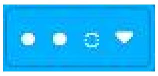
Programming

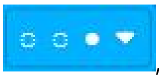
Directly import the hex file code we provided into the Makecode editor and download it to Micro:bit. If drag to build a new code, remember to add the Pybit extension.



•Add a smiley face icon to the "on start" block, stop the wheels, and set the battery type.
•In the "forever" block, read the values of the three sensors first, then insert the "if (judgment)" block five times.

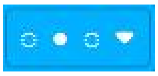
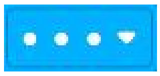
•First, determine whether the sensor status is , that is, the left sensor does not detect the black line, and the other two sensors detect the black line.
•Set the left wheel speed to 25 and the right wheel speed to 0, use the speed difference to make Pybit turn right and then return to the black line.

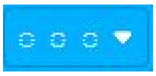
•Then determine whether the sensor status is  , that is, the right sensor does not detect the black line, and the other two sensors detect the black line.
• Set the right wheel speed to 25 and the left wheel speed to 0, and use the speed difference to make Pybit turn left and return to the black line

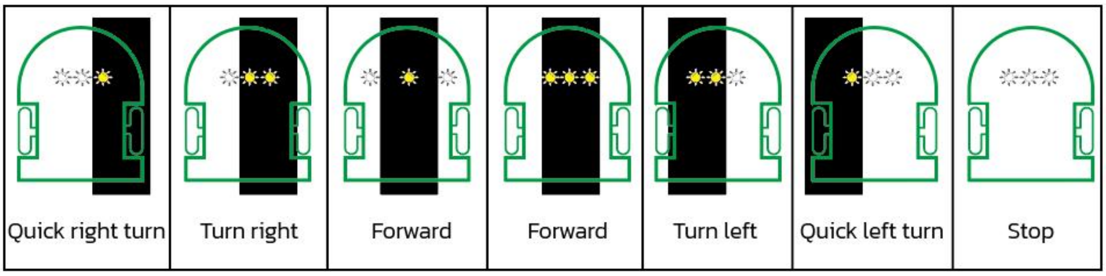
•First, determine whether the sensor status is , that is, the right sensor detects the black line, and the other two sensors do not detect the black line.
•Set the left wheel speed to 40 and the right wheel speed to 0, and use the speed difference to make Pybit turn right and return to the black line.

•Then determine if the sensor state is  , that is, the left sensor detects the black line, and the other two sensors do not detect the black line.

•Set the right wheel speed to 40 and the left wheel speed to 0, and use the speed difference to make Pybit turn left and return to the black line.

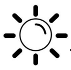
•Then determine if the sensor is in  and  states, which proves that the car is on the black line, and Pybit will go straight at a speed of 25.

•Then determine if the sensor is  , if it is, it proves that the car has moved away from the black line, and the car stops.

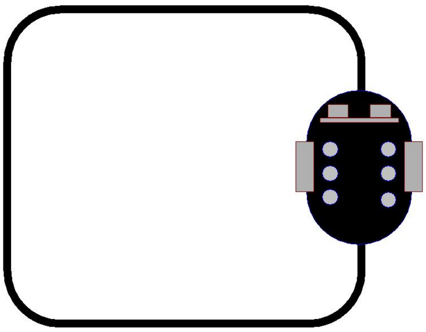


There are three LED indicators on the front of the robot car

 LED indicator lights up when black line is detected.

 The LED indicator turns off when no black line is detected.

Consequence



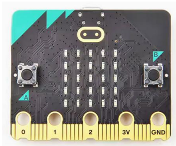
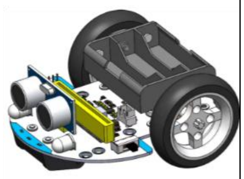
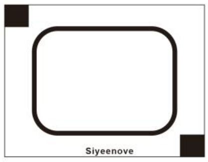
The Pybit will move forward at a constant speed following the black line on the map.

6.3 Don't cross the black line

Goal

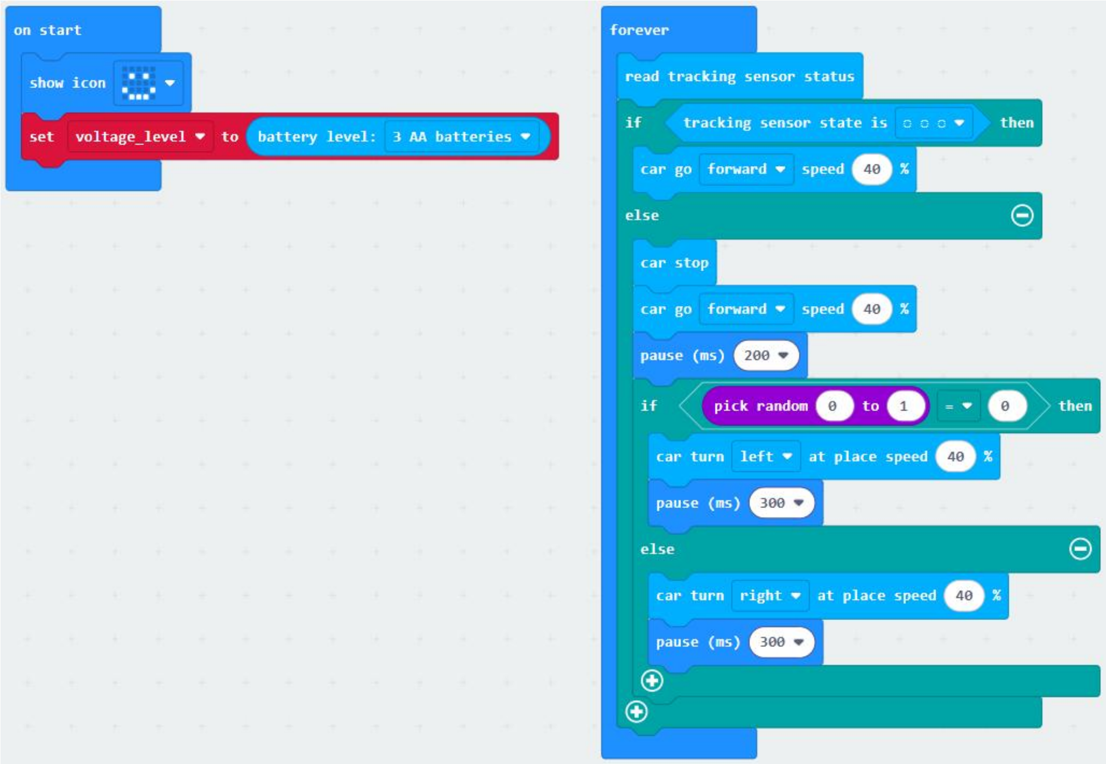
The Pybit is restricted to the area within the black line.

Things you need:

Computer	[Micro:bit v2.x.x]()	Micro USB cable	Pybit	Map
				

Programming

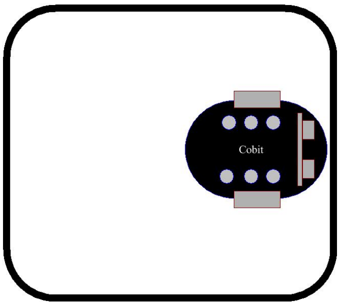
Directly import the hex file code we provided into the Makecode editor and download it to Micro:bit. If drag to build a new code, remember to add the Pybit extension.



•Add a smiley face icon to the "on start" block and set the battery type.
•First read the values of the three sensors in the "forever" block, and then insert the "if

(judgment)" block. If none of the three sensors detects a black line, the Pybit moves forward at full speed. Otherwise, the Pybit stops immediately, then reverses at a speed of 60, and then randomly turns left or right at a speed of 60.

Consequence



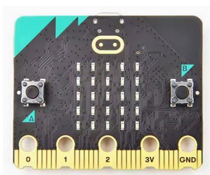
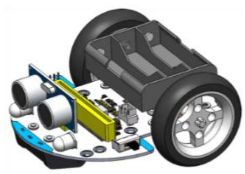
The Pybit will always run within the black line and will not cross the black line.

6.4 Follow your hand at a fixed distance

Goal

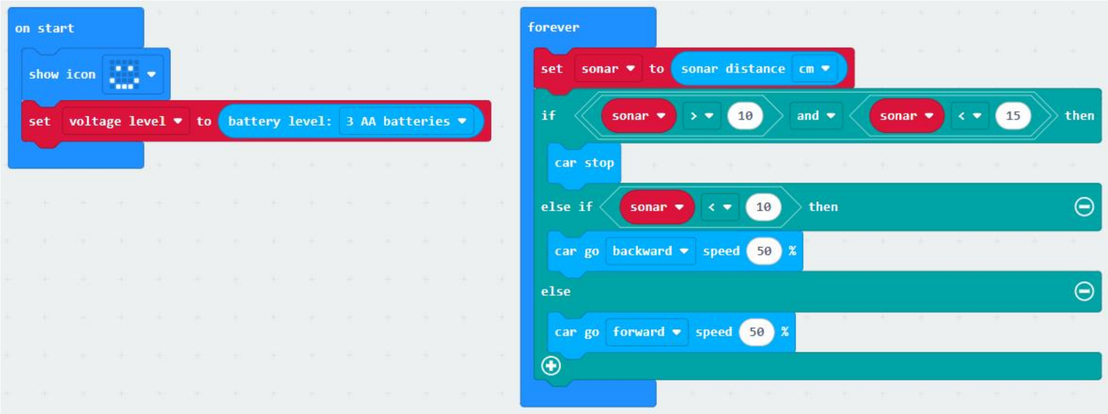
Let Pybit follow your hand at the same distance.

Things you need：

Computer	[Micro:bit v2.x.x]()	Micro USB cable	Pybit
			

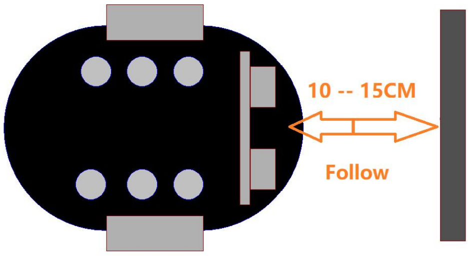
Programming

Directly import the hex file code we provided into the MakeCode editor and download it to Micro:bit. If drag to build a new code, remember to add the Pybit extension.



•Add a smiley face icon to the "on start" block and set the battery type.
•Set a sonar variable in the "forever" block to store the CM value returned by the ultrasonic module.
•When the ultrasonic module returns a value greater than 10 and less than 15, the car stops.
•When the return value of the ultrasonic module is less than 10, the robot is too close to the object and will move backwards.
•If neither, Pybit is too far from your hand, move forward to catch up with your hand and maintain proper position.

Consequence



When the Pybit is too far away from your hand, it will move forward. When it is too close, it will move backward. When the distance is right, it will stop.

Note

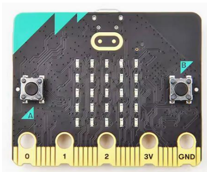
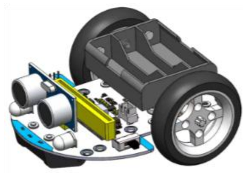
Q: When using Pybit, if you find that the car is running normally, but cannot run after connecting the ultrasonic sensor.
A: Please check whether the ultrasonic sensor is plugged into the wrong interface. It should be plugged into the sonar interface instead of the I2C interface behind.

6.5 Obstacle Avoidance

Goal

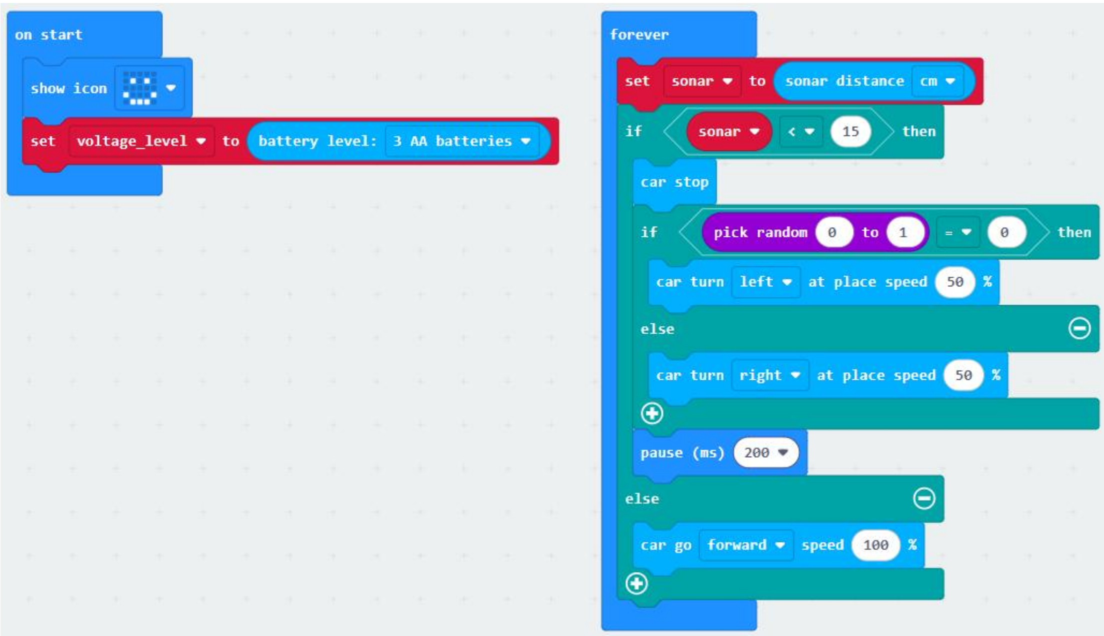
Let Pybit achieve autonomous driving and obstacle avoidance.

Things you need:

Computer	Micro:bit v2.x.x	Micro USB cable	Pybit
			

Programming

Directly import the hex file code we provided into the Makecode editor and download it to Micro:bit. If drag to build a new code, remember to add the Pybit extension.

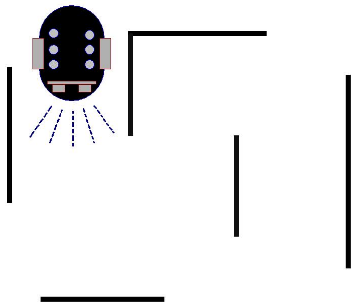


•Add a smiley face icon to the "on start" block and set the battery type.

•Set a sonar variable in the "forever" block to save the CM value returned by the ultrasonic module. When the return value of the ultrasonic module is less than 15, it proves that an obstacle has been detected 15CM ahead, and the car randomly turns left or right to avoid the obstacle.

•Else, Pybit goes full speed ahead.

Consequence



After the code is uploaded, the Pybit car moves forward at full speed. When it detects an obstacle within 15cm, it randomly rotates left or right by an angle and continues to move forward.

Note

Q: When using Pybit, if you find that the car is running normally, but cannot run after connecting the ultrasonic sensor.

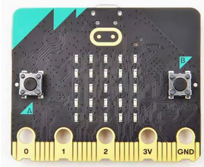
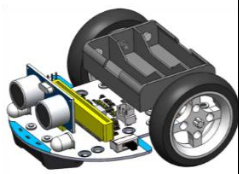
A: Please check whether the ultrasonic sensor is plugged into the wrong interface. It should be plugged into the sonar interface instead of the I2C interface behind.

6.6 Follow the Light

Goal

The dot matrix of the Micro:bit can sense the intensity of light. We use this function to allow Pybit to automatically find light sources in a dark environment.

Things you need:

Computer	Micro:bit v2.x.x	Micro USB cable	Pybit	flashlight
				

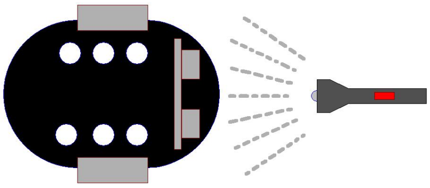
Programming

Directly import the hex file code we provided into the Makecode editor and download it to Micro:bit. If drag to build a new code, remember to add the Pybit extension.

• Set the battery type in the "on start" block.

• In the "forever" block, determine the brightness level. When the brightness level is less than the set threshold, the car turns left at 80% speed. When the brightness level is greater than the set threshold, the car moves forward at 80% speed.

Consequence



Shine a flashlight on the dot matrix of the Micro:bit. When the light level is detected to be less than 50, the car rotates in place. When the light level is detected to be greater than 50, the car moves forward at full speed.

Note

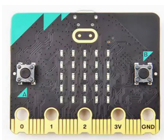
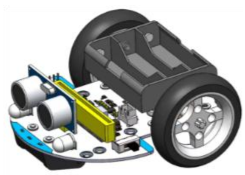
This may not work in brightly lit environments.

6.7 Infrared remote control

Goal

Use the infrared remote control to control the Pybit.

Things you need:

Computer	[Micro:bit v2.x.x]()	Micro USB cable	Pybit	Infrared remote control
				

Programming

Directly import the hex file code we provided into the Makecode editor and download it to Micro:bit. If drag to build a new code, remember to add the Pybit extension.

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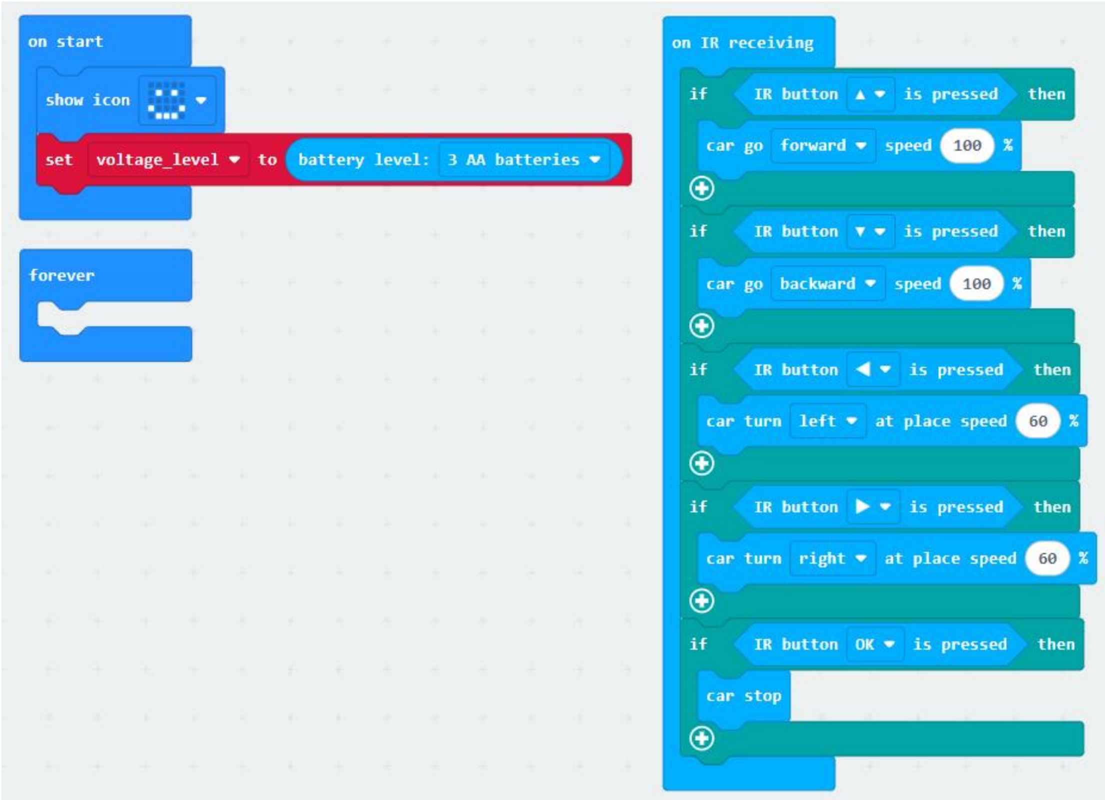


•Add a smiley face icon to the "on start" block and set the battery type.

•When Pybit receives an infrared signal, it determines which button is pressed; when the ▲ button on the infrared remote control is pressed, the car moves forward at full speed; when the ▼ button on the infrared remote control is pressed, the car reverses at full speed; when the ◀ button on the infrared remote control is pressed, the car turns left at full speed; when the ▶ button on the infrared remote control is pressed, the car turns right at full speed; when the OK button on the infrared remote control is pressed, the car stops immediately.

Consequence

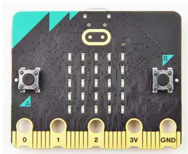
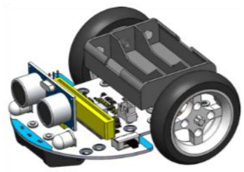
Use the infrared remote control to control the Pybit to move forward, backward, turn left, turn right, and stop.

6.8 Bluetooth APP control

Goal

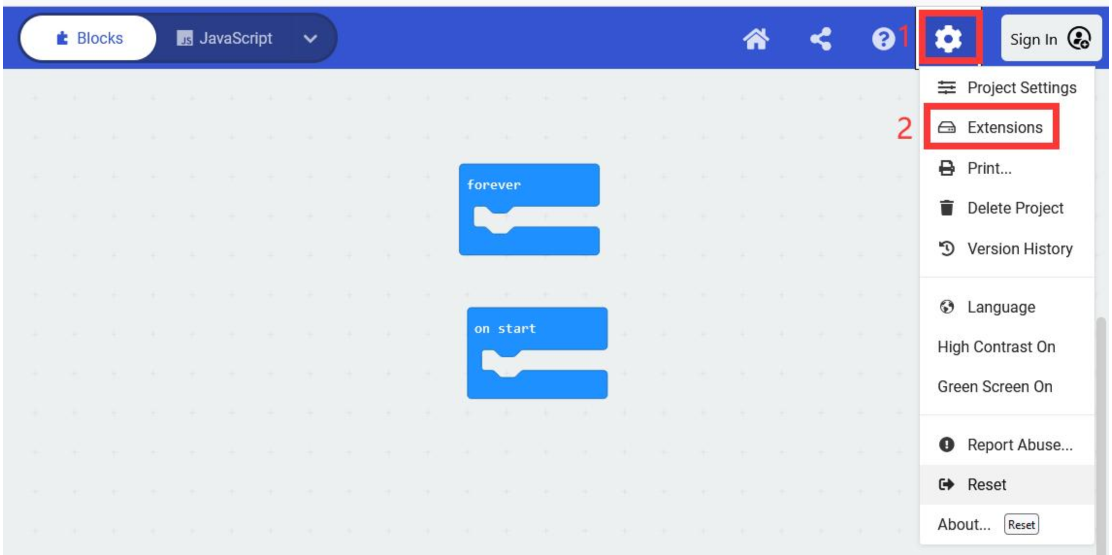
- Use Bluetooth APP of Android phone to control Pybit.
- Add Bluetooth extension

Things you need：

Computer	Micro:bit v2.x.x	Micro USB cable	Pybit	Android phone
				

6.8.1 Add Bluetooth expansion in Makecode

Search and add the Bluetooth extension:



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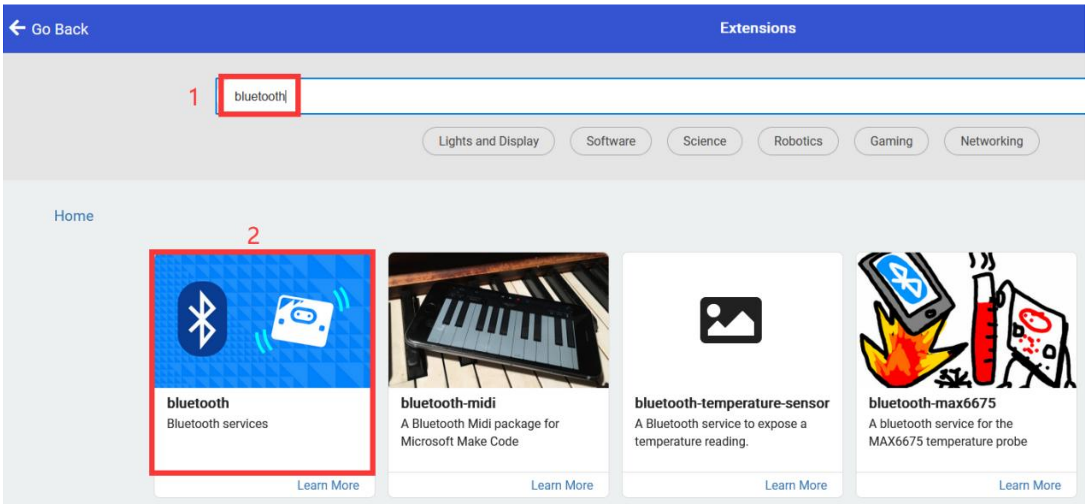
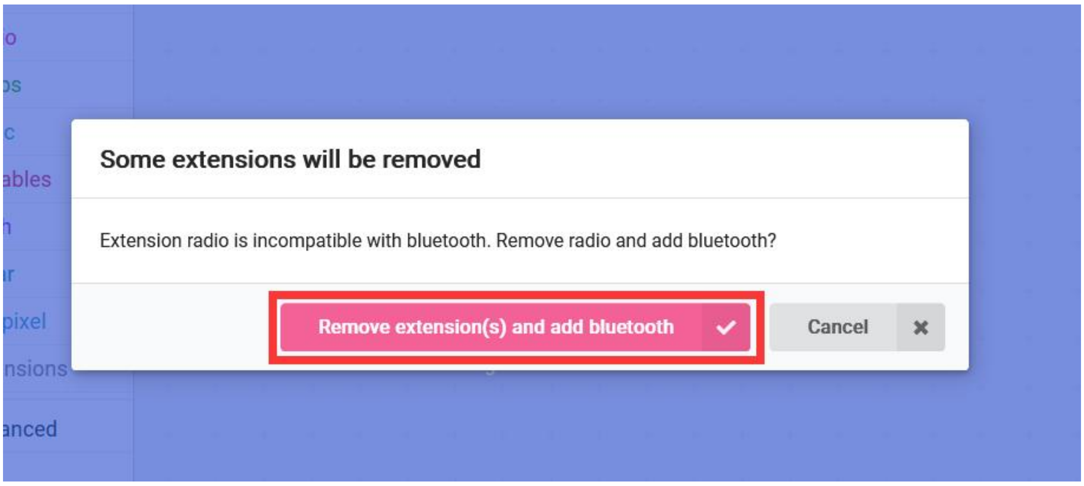
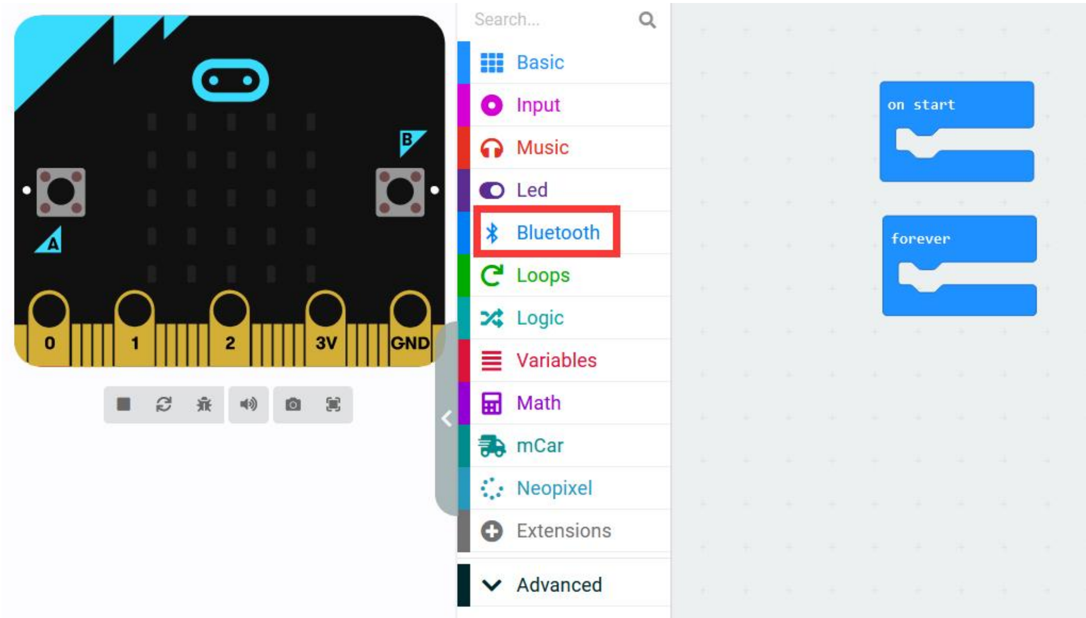


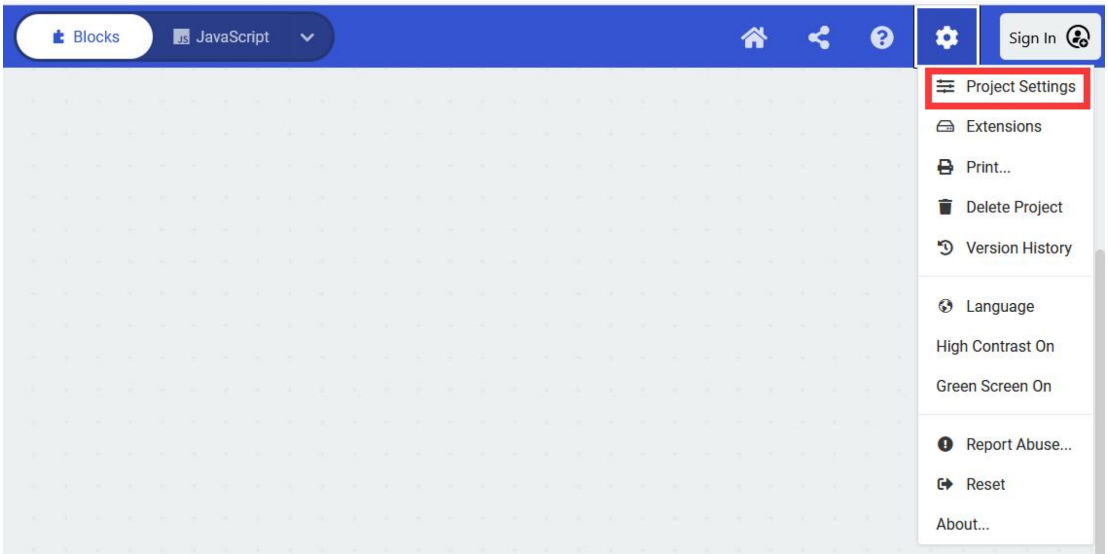
Extension radio is incompatible with bluetooth, Remove radio and add bluetooth.

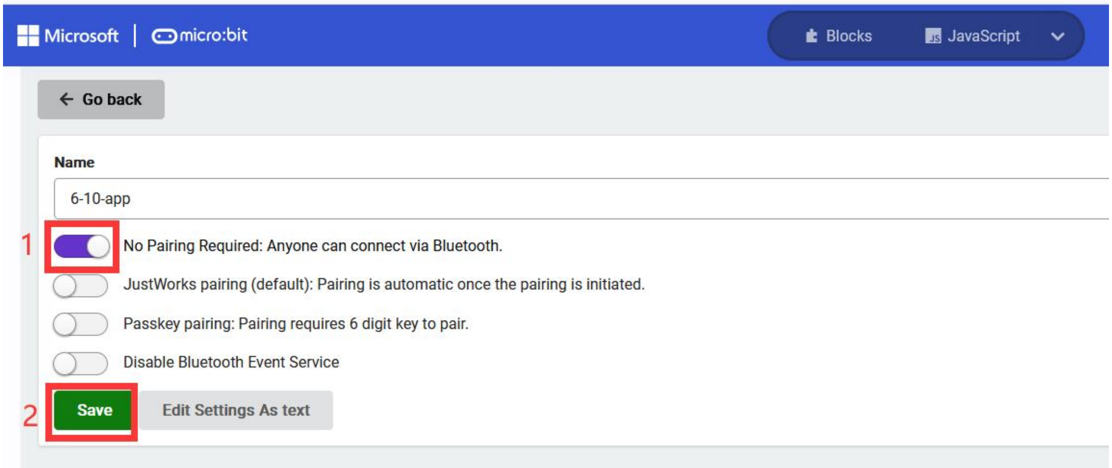


After successfully adding the Bluetooth extension, it will appear in the toolbox list.



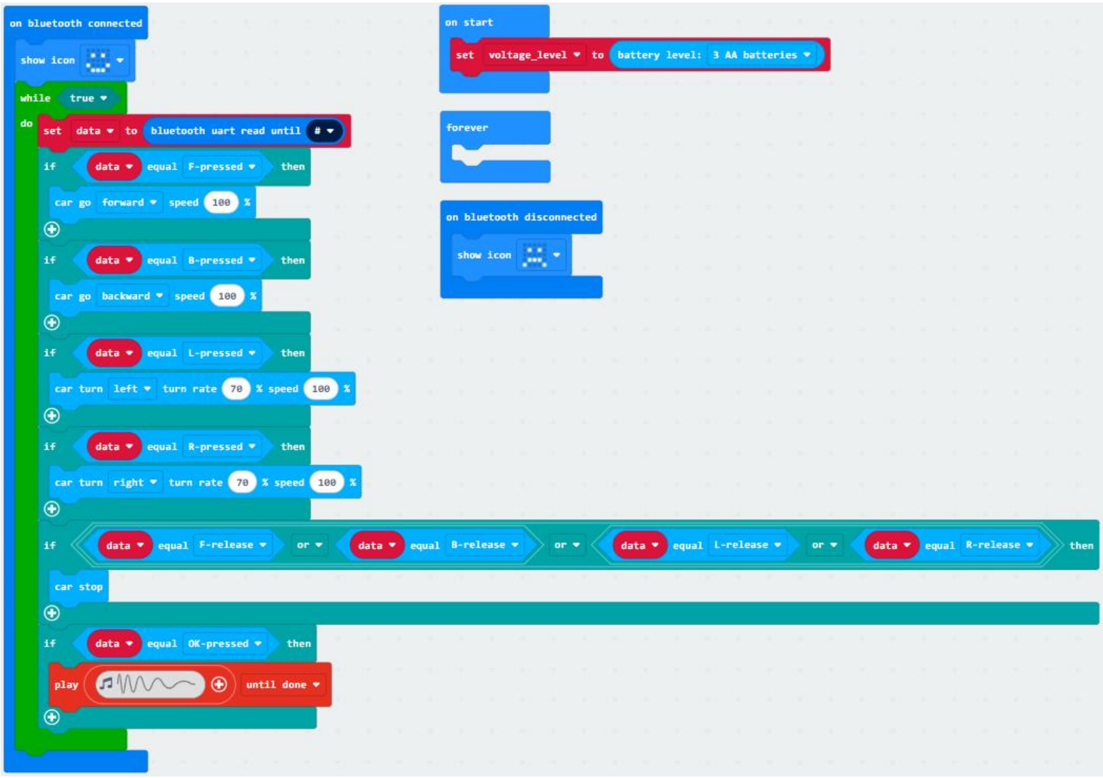
Set the micro:bit Bluetooth to a password-free connection:





6.8.2 Programming

Directly import the hex file code we provided into the MakeCode editor and download it to Micro:bit. If drag to build a new code, remember to add the Pybit extension.



Code Analysis:

•When successfully connected to the Bluetooth of Micro:bit, the 5*5 dot matrix displays a smiley face.

•Then enter an infinite loop.

•In the dotted loop, read the data received by the Micro:bit Bluetooth.

•If the "F" key value of the APP is received, the car moves forward.

•If the "B" key value of the APP is received, the car moves backward.

•If the "L" key value of the APP is received, the car turns left.

•If the "R" key value of the APP is received, the car turns right.

•When the pressed button is released, the car stops.

•If the "OK" key value of the APP is received, the buzzer sounds.

•When the Bluetooth of Micro:bit is disconnected, the 5*5 dot matrix displays a sad face.

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6.8.3 Install SIYEENOVE APP

In this section, we will guide you on how to download, install, and use the SIYEENOVE Bluetooth APP. You can also visit the official SIYEENOVE website to access it: https://siyeenove.com/app.

We provide both Android and iOS app versions for this robot.

(I) Introduction to the Android App

(II) Introduction to the iOS App

(Ⅰ) APP for Android Device

The following versions of Android system are supported:

Lowest supported version	Android 5.0（API 24）
Target fit version	Android 14（API 34）

Compatible with 80% + globally active Android devices.

For Android devices, we use Samsung's "Galaxy A54 5G" phone to demonstrate, with the following parameters:

<table>
<tr><td rowspan="4"></td><td>Product name</td><td>Galaxy A54 5G</td></tr>
<tr><td>Hardware version</td><td>REV0.4</td></tr>
<tr><td>Android version</td><td>13</td></tr>
<tr><td></td><td></td></tr>
</table>

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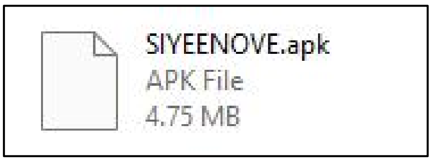
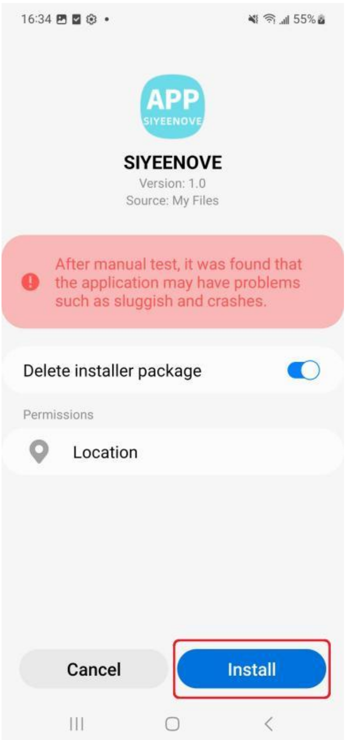
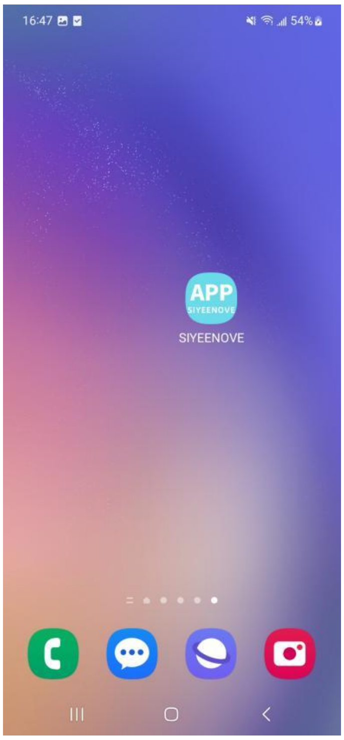
1) Install the SIYEENOVE Bluetooth APP for Android

The Android Bluetooth application is not published on the Google Play Store but is included in the tutorial package within the "Android APK" folder. There, you can copy the "SIYEENOVE.apk" to your phone, locate the APK in your phone's interface, and then tap on it to install.



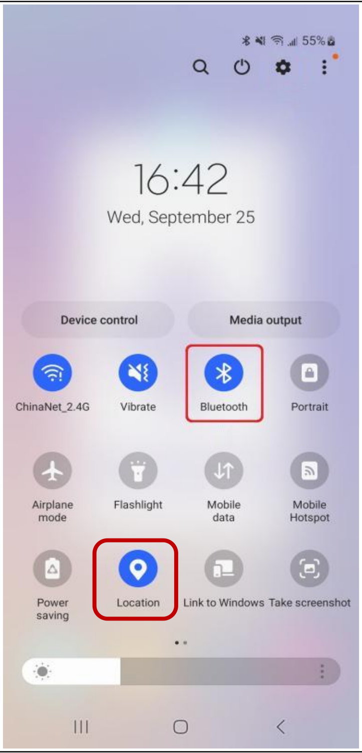
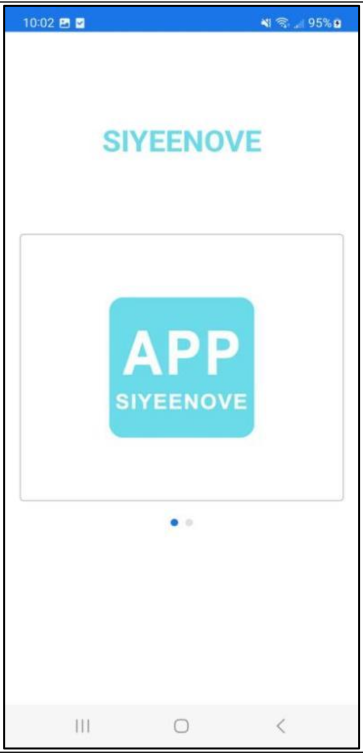
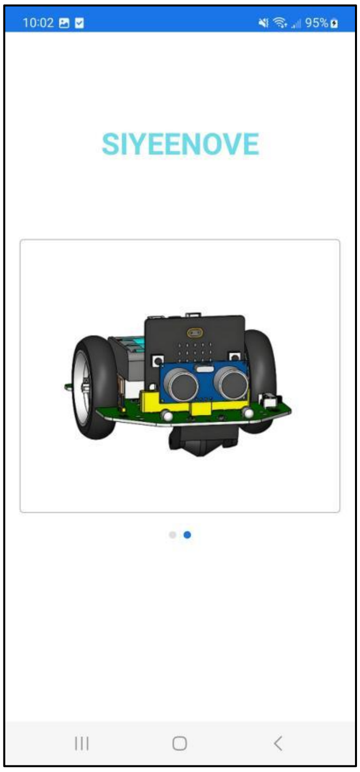
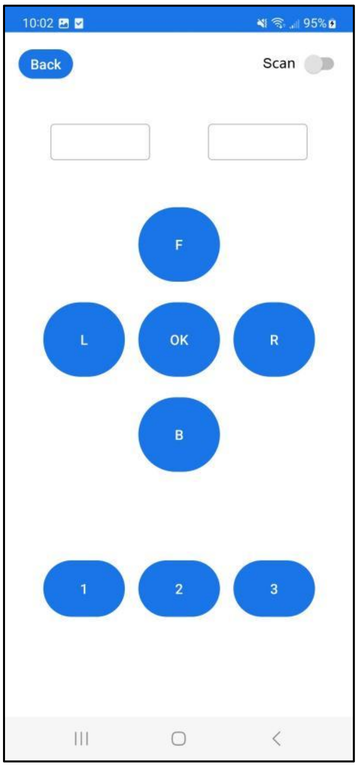
You can also download the Android app by scanning the QR code below using the scanning feature on your Android device.

If you encounter any security warnings during the download or installation of the app, please disregard them and proceed by clicking 'download' and 'Install'! We assure you that the app is safe and free from viruses.

<table>
<tr><th>Click to install</th><th>Installation completed</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

2) Start the APP

Steps：

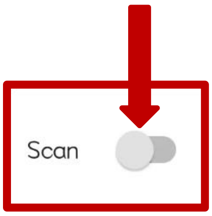
1. Turn on **Bluetooth and Location** Services on your phone.	2. Start the APP and enter the home page
	
3.Swipe left to access the Pybit interface.	4. Click on the Pybit image to homepage
	

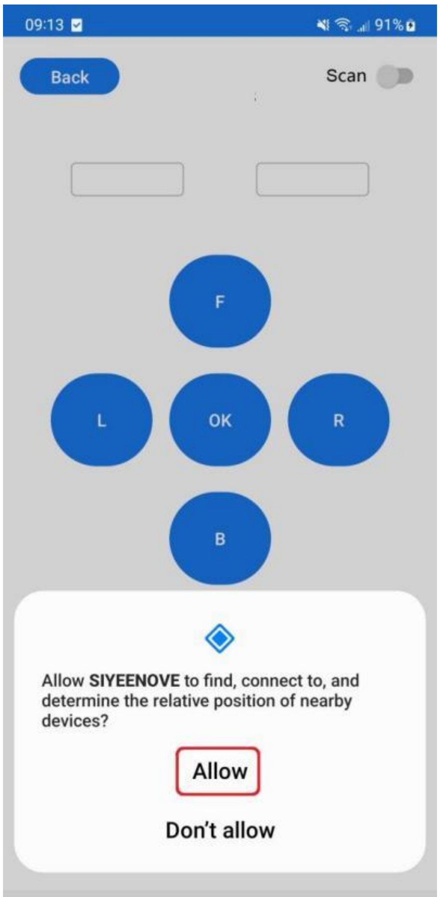
3) Connect to the Micro: bit with Bluetooth

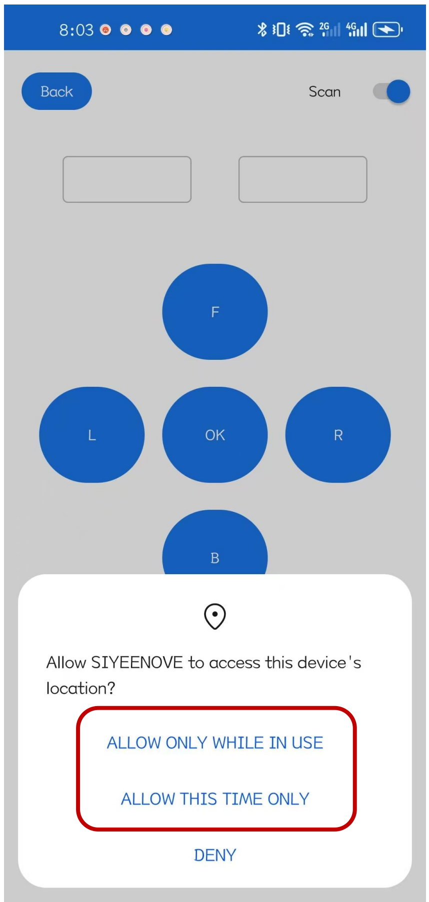
Warn:

To use the Bluetooth app on an Android device, you must enable the "**Location**" or "**Nearby Devices**" feature to allow it to search for nearby devices.

When you tap the "**Scan**" button in the top-right corner of the Pybit control page to start a Bluetooth scan, if the "Location" or "Nearby Devices" permission is not enabled, the app will prompt you to enable it. Be sure to click "**Allow**," otherwise the app will not work.







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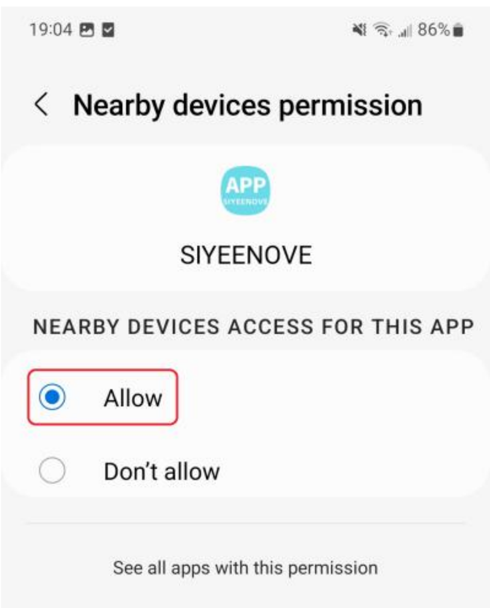
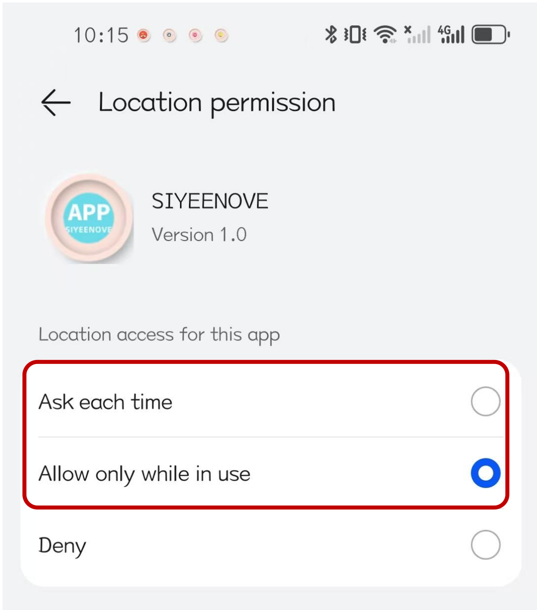
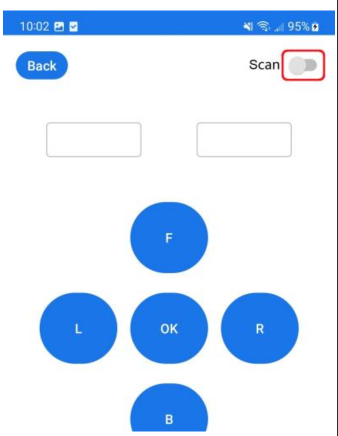
If you accidentally click "**Don’t allow**" or "**Deny**," the app will be unable to search for Bluetooth devices and will display a pop-up message saying, "**Permission is permanently denied**."
In this case, you need to click "**Go to Settings**" to manually enable the "Location" or "Nearby Devices" permission for the app.

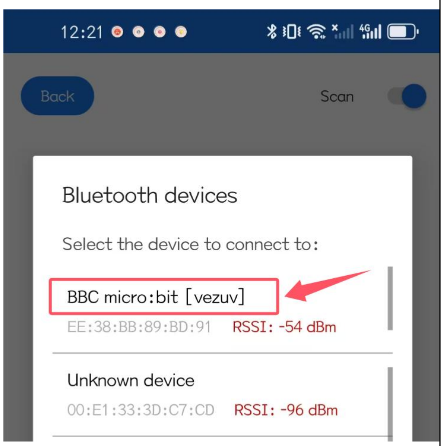


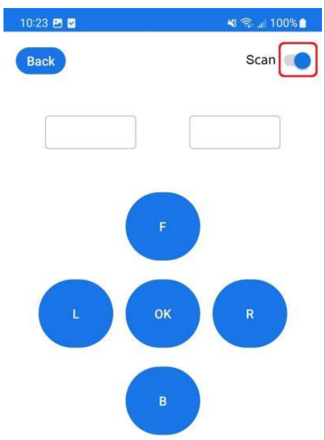


Note: The code “**microbit-6-10-app**” must be uploaded to the micro:bit before you can search for and connect to the Micro:bit device via the SIYEENOVE APP.

1.Scan	2.Select micro:bit	3.Successfully connected







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(Ⅱ) APP for iOS Device

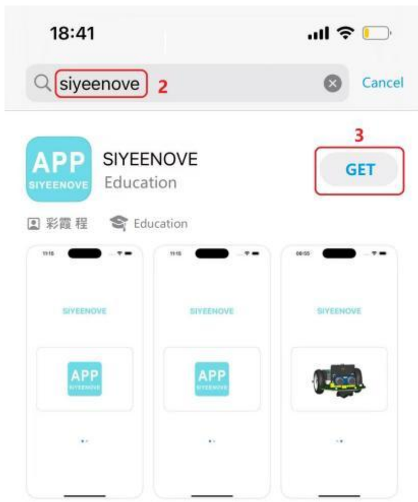
The iOS APP supports:

Minimum supported version:	15.6

Here, we are using the "iPhone 14" for demonstration, with the specific parameters as follows:

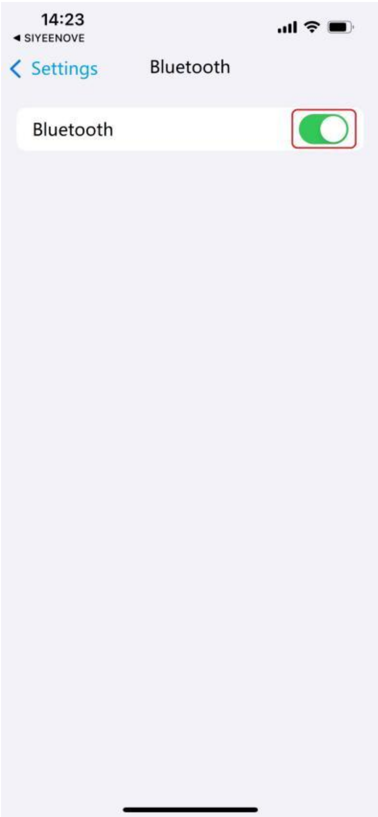
	Product name	iPhone 14
	Software version	17.6.1

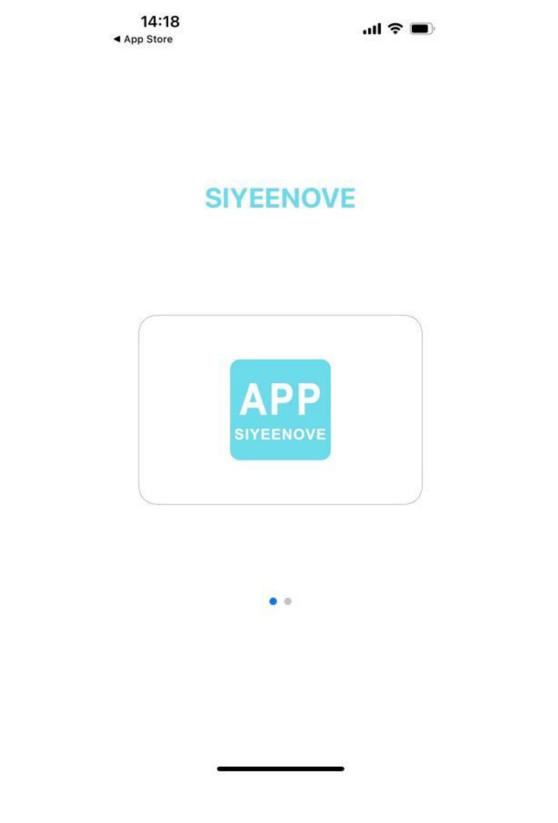
1) Install the SIYEENOVE Bluetooth APP for iOS

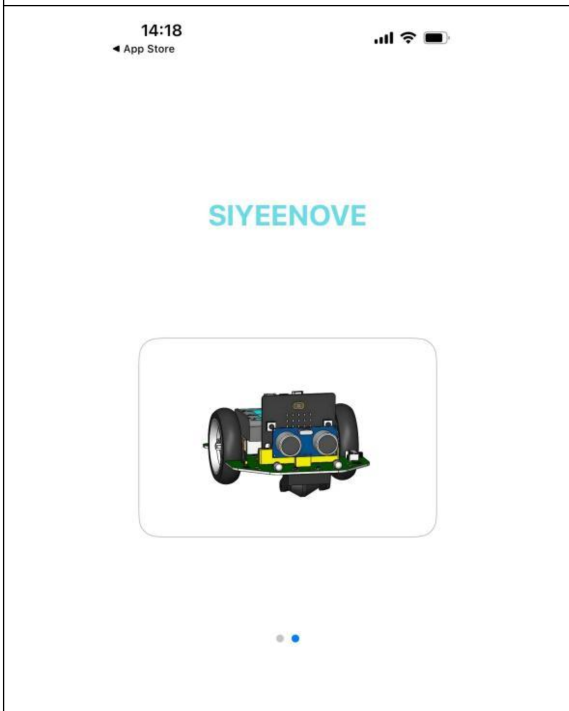
You can search for the app in the Apple Store using keywords such as "siyeenove," "SIYEENOVE," "Siyeenove," "SiyeenoveApp," or "siyeenoveapp," and then proceed to install it.	

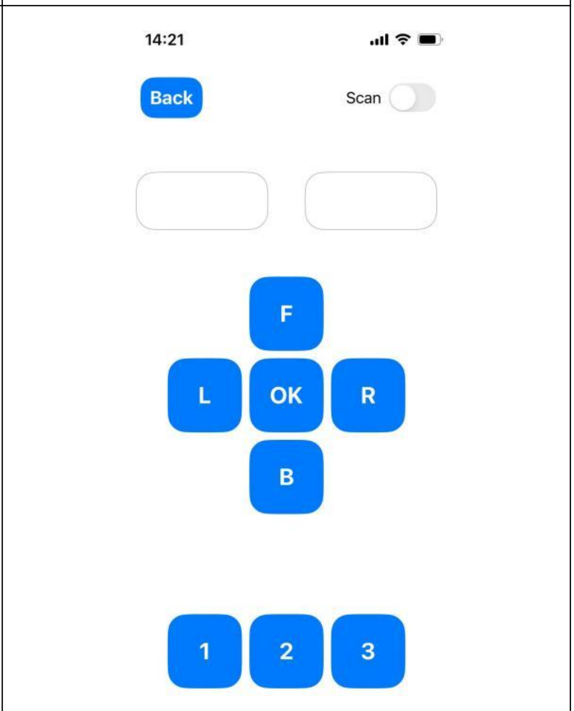
2) Start the APP

Steps：

<table>
<tr><td>1. Turn on Bluetooth Services on your phone.</td><td>2. Start the APP and enter the home page</td></tr>
<tr><td>


</td><td>


</td></tr>
<tr><td>3.Swipe left to access the Pybit interface.</td><td>4. Click on the Pybit image to homepage</td></tr>
<tr><td>


</td><td>


</td></tr>
</table>

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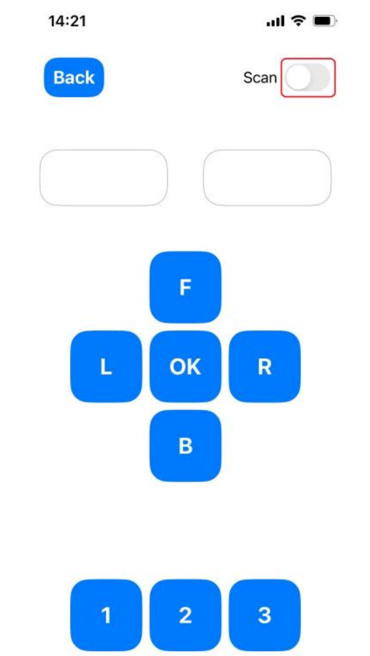
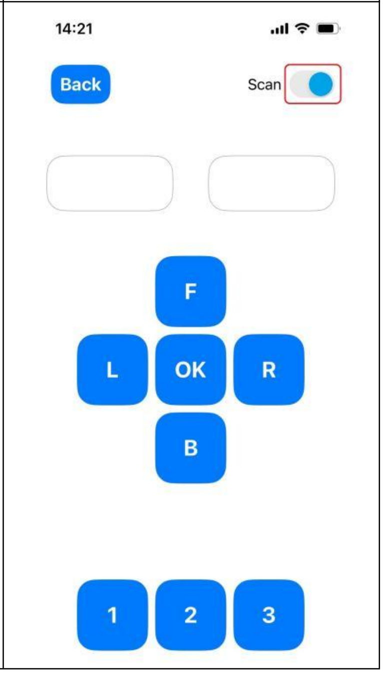
3) Connect to the Micro: bit with Bluetooth

Note: The code "**microbit-6-10-app**" must be uploaded to the micro:bit before you can search for and connect to the Micro:bit device via the SIYEENOVE APP.

1.Scan	2.Select micro:bit	3.Successfully connected
		

APP Instructions

Sending Instructions		
Icon	Action	APP Sent Character Command String
F	Press	$-6-1-0-1-#
F	Release	$-6-1-0-0-#
OK	Press	$-6-1-1-1-#
OK	Release	$-6-1-1-0-#
B	Press	$-6-1-2-1-#
B	Release	$-6-1-2-0-#
L	Press	$-6-1-3-1-#
L	Release	$-6-1-3-0-#

R	Press	$-6-1-4-1-#
R	Release	$-6-1-4-0-#
1	Press	$-6-1-5-1-#
1	Release	$-6-1-5-0-#
2	Press	$-6-1-6-1-#
2	Release	$-6-1-6-0-#
3	Press	$-6-1-7-1-#
3	Release	$-6-1-7-0-#

Receiving Instructions		
Display Box	Position	APP Received Character Command String
	Left	$-6-1-8-xxx-#
	Right	$-6-1-9-xxx-#

Note: xxx represents any character except $, -, and #.

Consequence

After the APP and micro:bit Bluetooth are successfully connected, you can use the App to control the Pybit, below are the instructions corresponding to the app buttons

Button	F	B	L	R	OK
Function	Press>forward Release>stop	Press>backward Release>stop	Press>turn left Release>stop	Press>turn right Release>stop	Press>buzzer sounds

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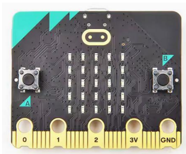
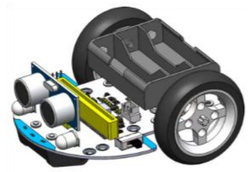
6.9 Bluetooth APP Plus control

Goal

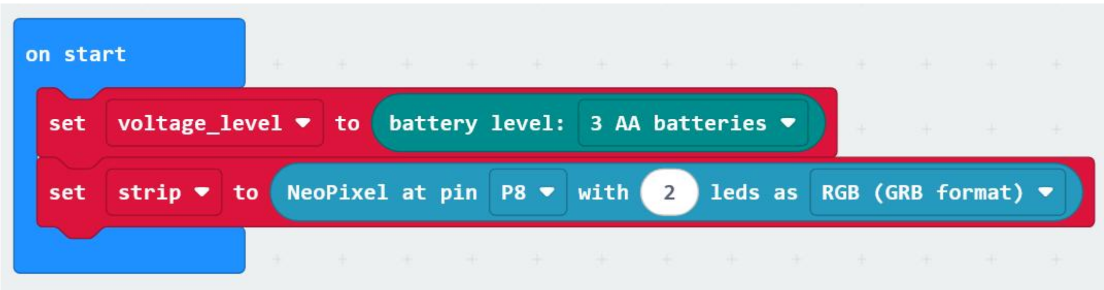
- Use Bluetooth APP of Android phone to control Pybit.
- Add more app control functions.

Things you need:

Computer	Micro:bit v2.x.x	Micro USB cable	Pybit	Phone
				

Programming

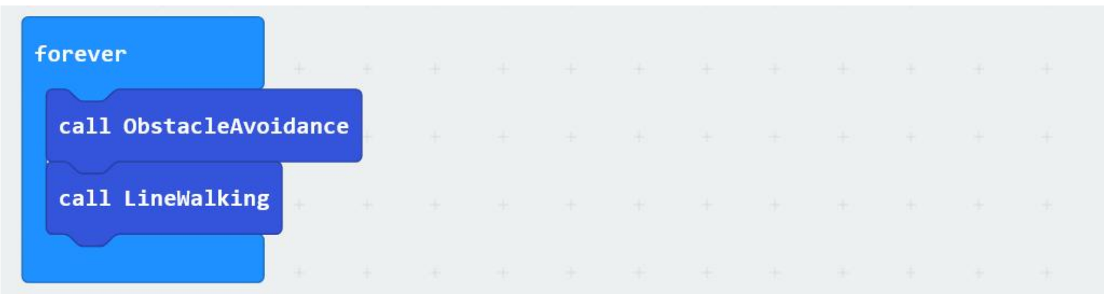
1. Micro:bit startup code:



•Set Pybit to use 3 AA batteries.

•Set up Pybit to control 2 RGB ambient lights using the P8 pin of the micro: bit.

2. Micro:bit loop code after startup:



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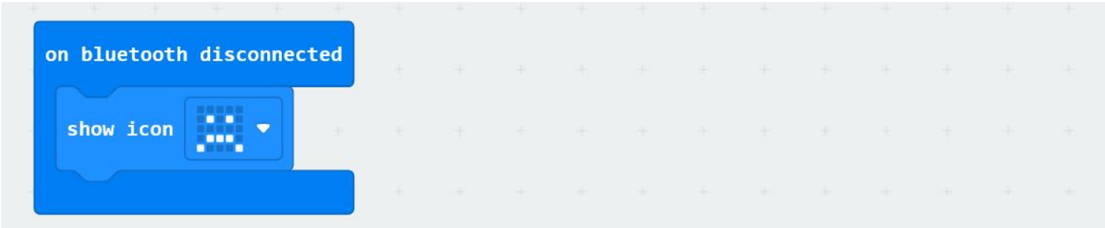
•Execute obstacle avoidance function.

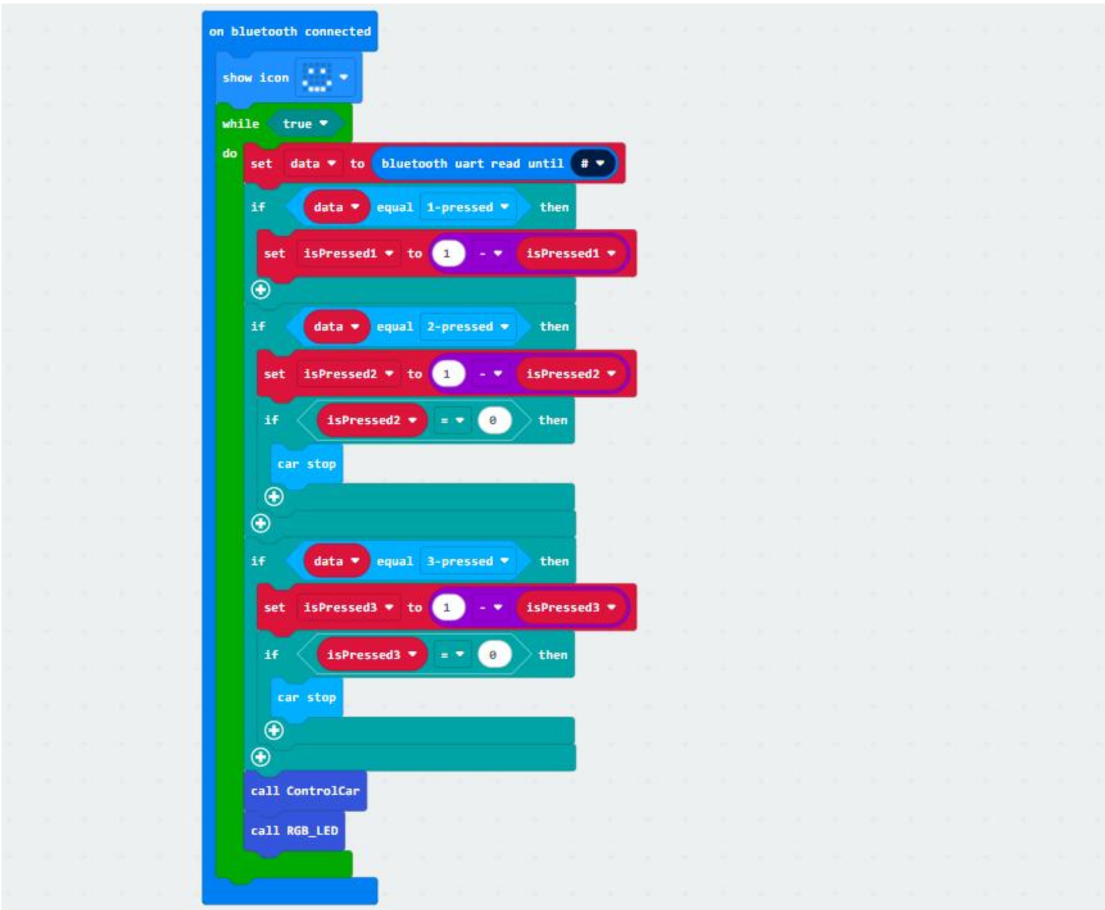
•Execute the detection line function.

3. Bluetooth disconnection code



•When Bluetooth is disconnected, the micro: bit dot matrix displays a sad face.

4. Bluetooth successful connection code



•When Bluetooth is successfully connected, the micro: bit dot matrix displays a smiling face.

•Receive APP instructions in a while loop.

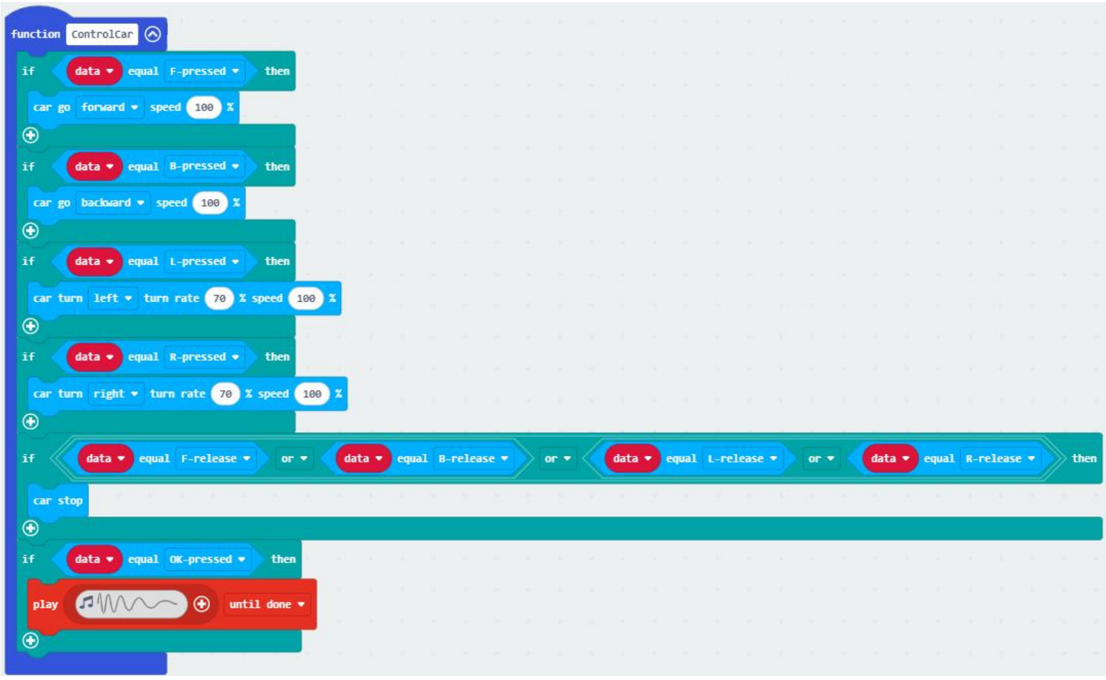
•If button 1 is pressed, change the value of isPressed1 to either 0 or 1.

•If button 2 is pressed, change the value of isPressed2 to 0 or 1. If it is equal to 0, Pybit stops.

•If the 3 keys are pressed, change the value of isPressed3 to 0 or 1. If it is equal to 0, Pybit stops.

- Execute control car functions
- Execute RGB_LED function

5. Control Pybit function



•If the F button is pressed, Pybit runs forward at 100% speed.

•If the B button is pressed, Pybit runs backwards at 100% speed.

•If the L button is pressed, the Pybit turns left at 70% turn rate and 100% speed.

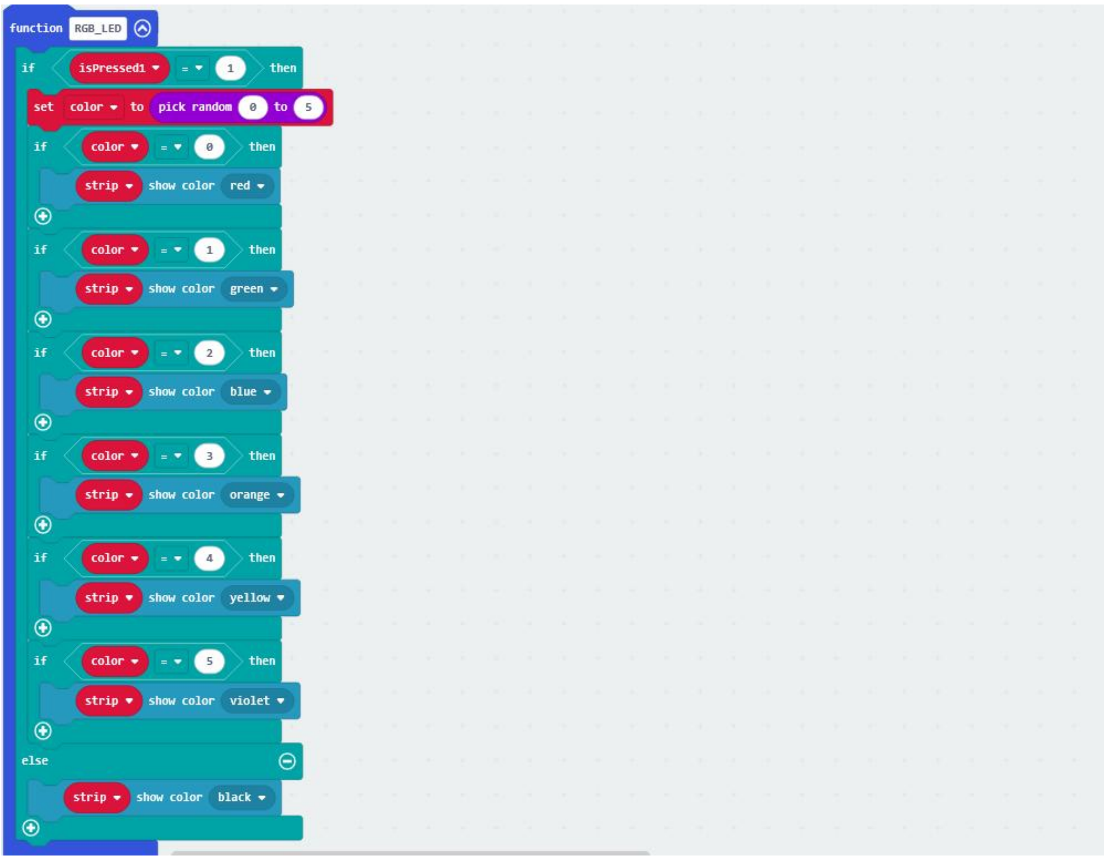
•If the R button is pressed, the Pybit turns right at 70% turn rate and 100% speed.

•If the F and B buttons are released, Pybit will stop.

•If the L and R buttons are released, Pybit will stop.

•If the OK button is pressed, Pybit will beep.

6. RGB_LED ambient light function



•If the isPressed1 variable is equal to 1, determine which color the light is on.

•Randomly generate any number between 0-5.

•The random number is 0, and the ambient light is red.

•The random number is 1, and the ambient light is green.

•The random number is 2, and the ambient light is blue.

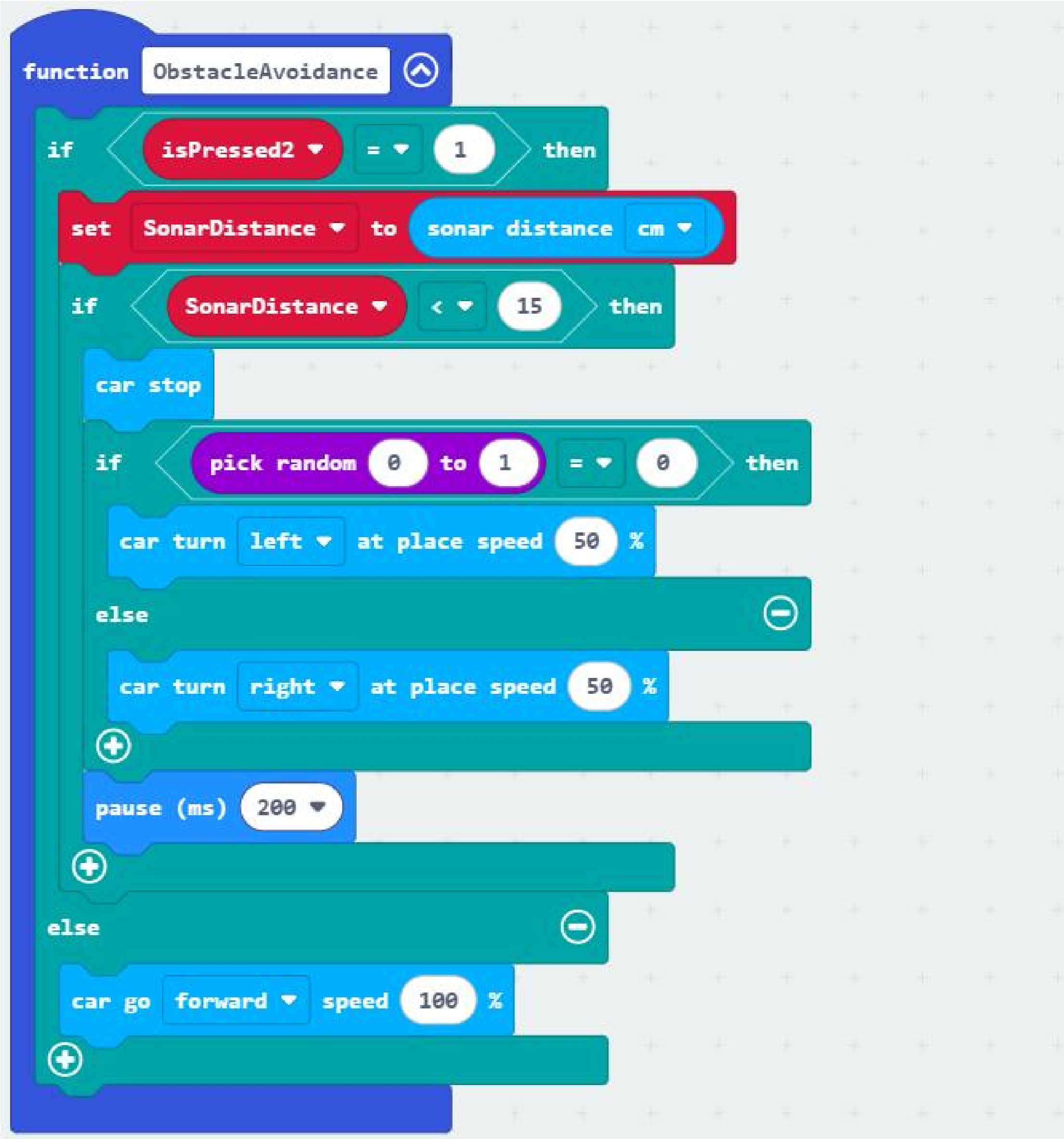
•The random number is 3, and the ambient light is orange.

•The random number is 4, and the ambient light is yellow.

•The random number is 5, and the ambient light is violet colored.

•If the isPressed1 variable is not equal to 1, turn off the light.

7. Obstacle avoidance function



•If the isPressed2 variable is equal to 1, measure the distance and then avoid obstacles.

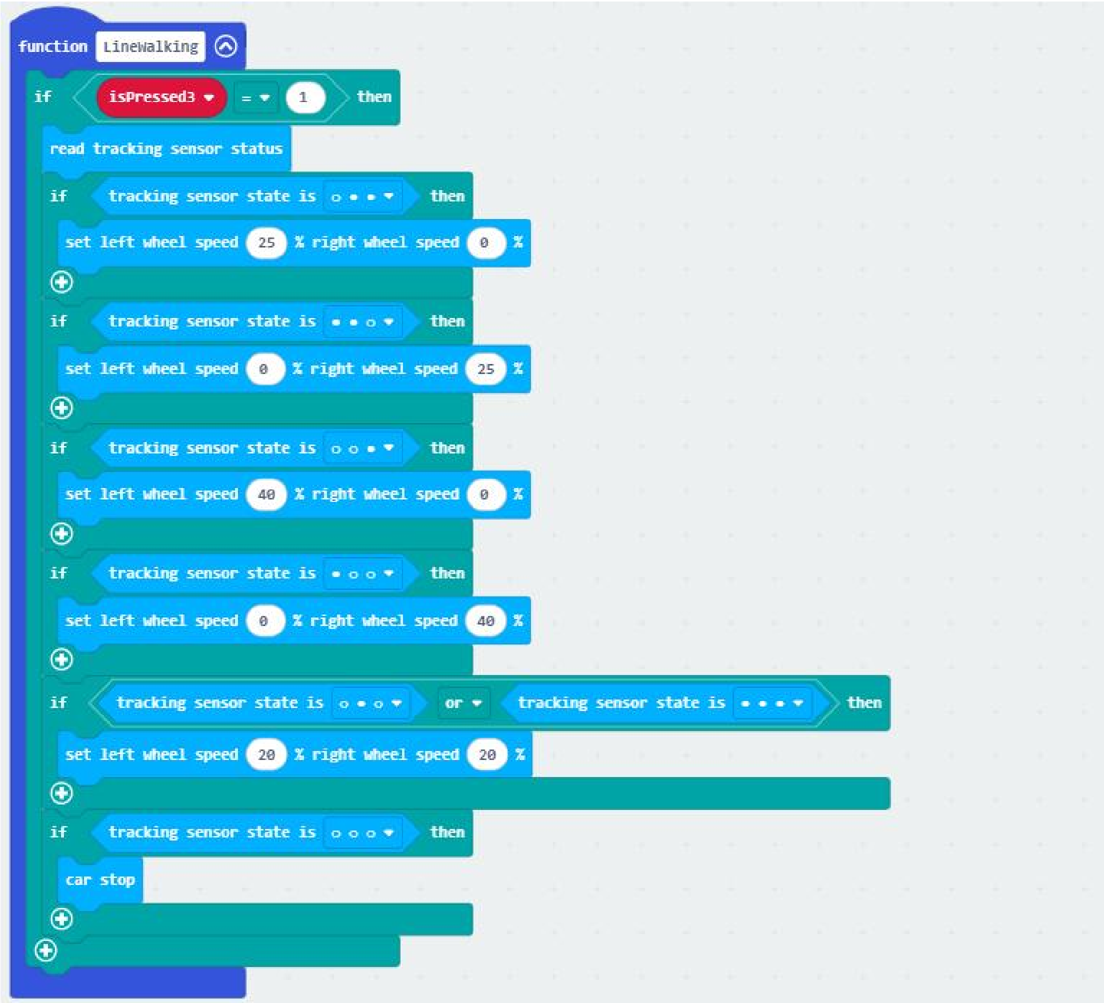
Set a SonarDistance variable to store the values returned by ultrasound waves.

•When the ultrasound return value is less than 15CM, it proves that an obstacle has been detected 15CM ahead, and Pybit randomly turns left or right.

• Delay of 200ms

•If the ultrasonic return value is greater than or equal to 15CM, Pybit will move forward at full speed.

8. Line checking function



If the isPressed3 variable is equal to 1, enter the line checking function, and the code function is the same as in section 6.2.

Install SIYEENOVE Bluetooth APP

If you haven't installed it yet, please refer to the previous section to download and install it.

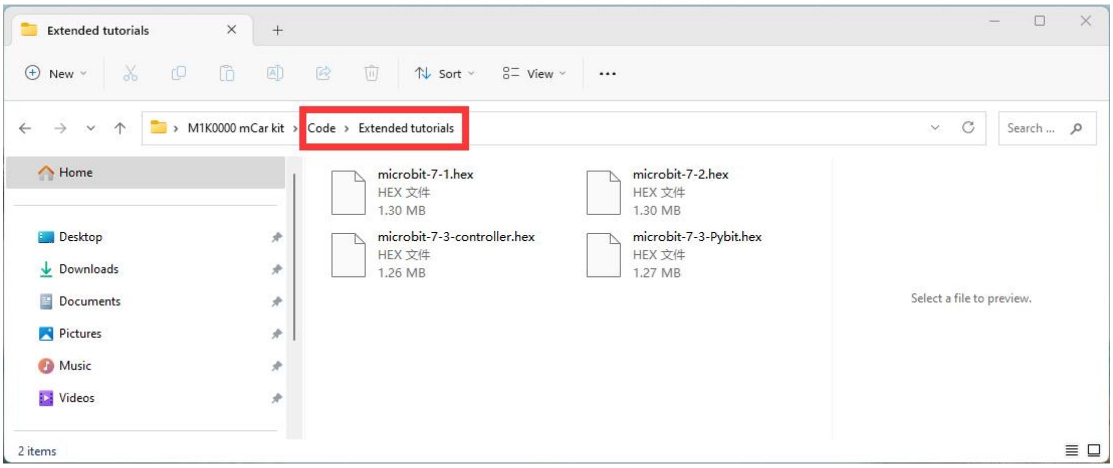
Consequence

After the APP and micro:bit Bluetooth are successfully connected, you can use the App to control the Pybit, below are the instructions corresponding to the app buttons:

Button Function Table				
Button	**F**	**B**	**L**	**R**
Action	Press → Move forward	Press → Move backward	Press → Turn left	Press → Turn right
	Release → Stop	Release → Stop	Release → Stop	Release → Stop
Button	**OK**	**1**	**2**	**3**
Action	Press → Activate buzzer	Toggle ambient light	Toggle obstacle avoidance	Toggle line tracking

7. Pybit Extended Tutorial

The sample codes for the extended tutorials are all saved in the "Code -> Extended tutorials" file:



Reminder!!!

You don't need to add the Pybit extension if you import and use the code we provided. ---> Recommend!

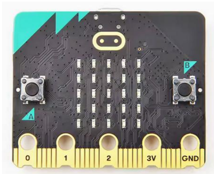
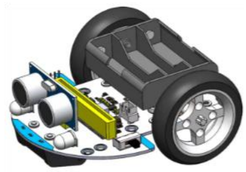
If you create a new project and drag and drop blocks to build code, you need to manually add the Pybit extension, please refer to sections 3 and 4.1.

7.1 Drive a 180 degree servo

Goal

Learn how to drive a 180-degree Lego servo, and get some inspiration for expanding the capabilities of your Pybit.

Things you need:

Computer	Micro:bit v2.x.x	Micro USB cable	Pybit	180° Lego servo
				

Note: The kit does not include a 180-degree Lego servo. You will need to prepare it yourself.

Programming

Directly import the hex file code we provided into the Makecode editor and download it to Micro:bit. If drag to build a new code, remember to add the Pybit extension.

Consequence

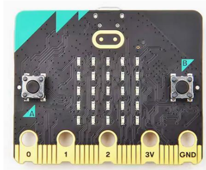
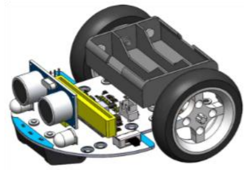
The 180 degree Lego servo shaft rotates cyclically between 0 and 180 degrees.

7.2 Drive a 360 degree servo

Goal

Let the 360-degree Lego servo rotate evenly and rapidly in the positive direction.

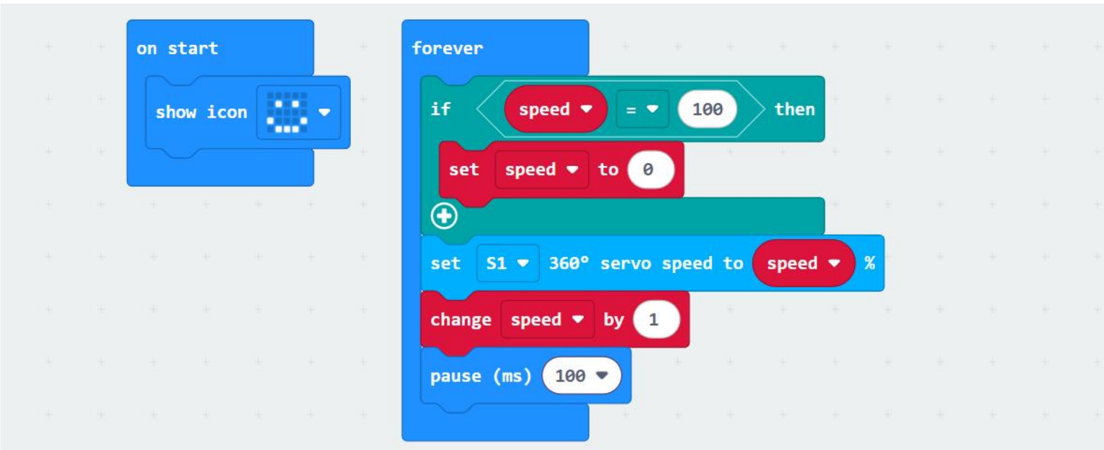
Things you need:

Computer	Micro:bit v2.x.x	Micro USB cable	Pybit	360° Lego servo
				

Note: The kit does not include a 360-degree Lego servo.

Programming

Directly import the hex file code we provided into the Makecode editor and download it to Micro:bit. If drag to build a new code, remember to add the Pybit extension.



Consequence

The 360-degree Lego servo rotate evenly and rapidly in the positive direction.

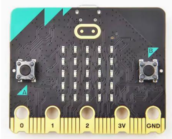
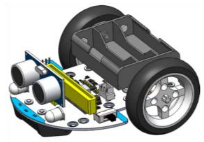
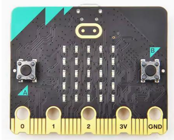
7.3 Accelerometer wireless control

Note: This is an optional project, the kit does not include any Micro:bit, if you have two Micro:bit, you can try it.

Goal

- Use the accelerometer of another Micro:bit to wirelessly control Pybit, which can control the 360-degree direction and speed of the car.
- Both Micro:bits need to upload programs.

Things you need：

Computer	Micro:bit v2.x.x	Micro USB cable	Pybit	Another micro:bit V2
				

Warning! If you cannot find the "Radio" category on the left side in Makecode：

You can add it through "Extensions".

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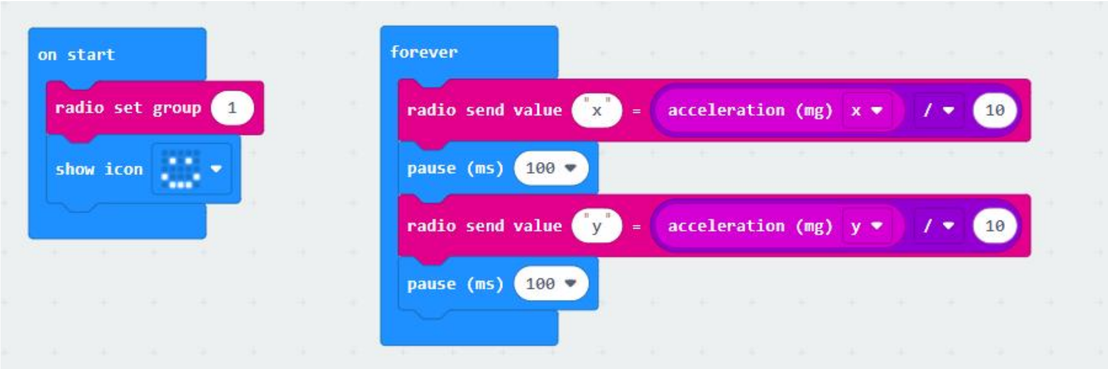
Programming

-Upload code to the another Micro:bit as a remote controller

Directly import the hex file code we provided into the MakeCode editor and download it to Micro:bit. If drag to build a new code, remember to add the Pybit extension.



• Set the radio group to 1 in the "on start" block;

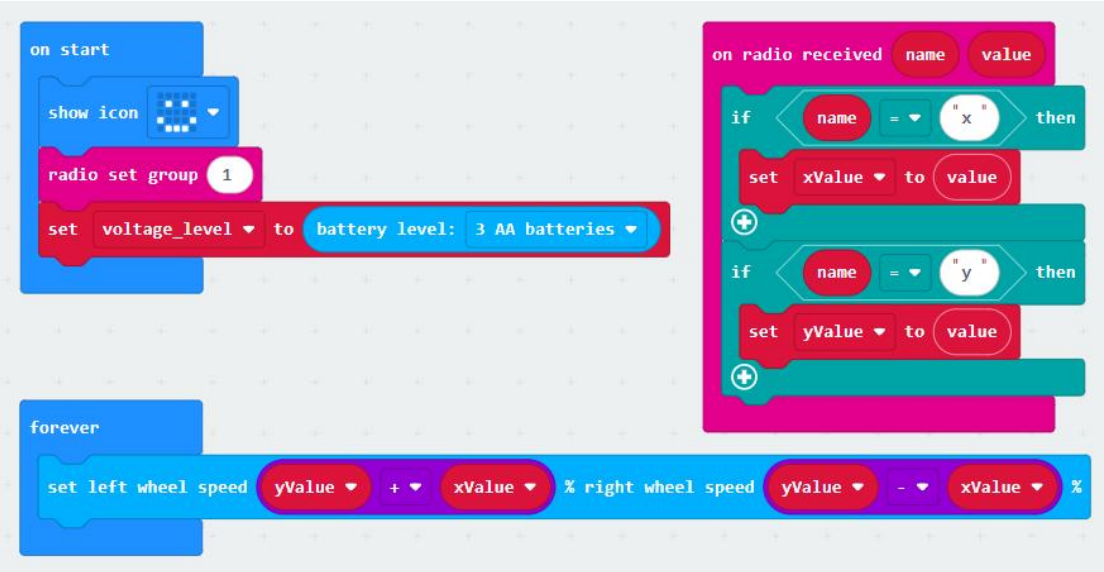
•In the "forever" block, the value of radio send value x is the x-axis acceleration value divided by 10.

•In the "forever" block, the value of radio send value y is the y-axis acceleration value divided by 10.

•Because the acceleration value range is -1024~0~1024, after dividing by 10 it can be approximated as -100~0~100.

-Upload code to Microbit for Pybit

Directly import the hex file code we provided into the MakeCode editor and download it to Micro:bit. If drag to build a new code, remember to add the Pybit extension.



•Add a smiley face to the "on start" block; set the radio group to 1. Make sure it is the same group as the remote control, otherwise they will not match; then set the battery type.

•Then insert two judgment statements in the "radio received" block to judge whether the radio received value name is x or y;

•When the name value received by the radio is x, it is the accelerometer X-axis data, and the value is saved to the xValue variable;

•When the name value received by the radio is y, it is the accelerometer Y-axis data, and the value is saved to the yValue variable;

•In the "forever" building block, set the left wheel speed to yValue+xValue and the right wheel speed to yValue-xValue.

Consequence

▸ Tilt the micro:bit on the control end in different directions to control the Pybit to move in different directions.

▸ Use the tilt angle of the micro:bit on the control end to control the speed of the Pybit

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8.QA

8.1 Unable to upload code to Micro:bit

☛Do you use a USB cable with data communication function?

☛Does the USB cable connect well?

8.2 The Micro:bit disk name is not displayed correctly

☛If the disk name is displayed as: MAINTENANCE (Normal is MICROBIT), it means that it has entered the firmware update mode. If you press and hold the micro:bit reset button and then power on the micro:bit, the micro:bit will enter the firmware update mode. At this time, we can update the firmware and the micro:bit disk name will return to normal. You can refer to the following link:
https://microbit.org/get-started/user-guide/firmware/

8.3 After uploading a new code, the motor still rotates

☛ This is because the car stop block is not called again:

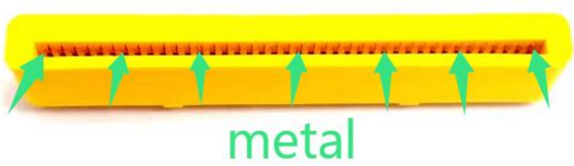
Car stop	Set all ▾ wheel to stop

8.4 Neither the motor nor the headlights work

☛Make sure the micro:bit is fully plugged into the Pybit slot.

☛Make sure the micro:bit Edge Connectors are clean.

☛Make sure the Pybit slots are clean.



8.5 Other issues

☛Please check if the assembly steps are correct?
☛Please check if the battery power is sufficient?
☛Please check if the battery model used is correct?

9.Contact us

If you couldn't find a solution above, please contact our support team.

To help us assist you quickly, please have the following information ready:

Your order number.

Product model (e.g., M1C0001) and software version (e.g., MakeCode v3.0.XX).

A detailed description of the issue or your question.

Steps you have already tried.

Any relevant error message screenshots, photos, or code snippets.

Other Inquiries?

We value your input and are always looking to improve. Please feel free to reach out for:

Tutorial Errors & Feedback: Help us make our documentation better.

Product Ideas & Suggestions: We'd love to hear your great ideas.

Partnerships & Collaboration: Interested in working together? Let's talk.

Discounts & Promotions: Inquire about educational or bulk pricing.

Anything Else: For all other non-technical questions.

Support Channels

support@siyeenove.com

https://siyeenove.com

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